

Statistical Data Analysis and Supplement Information

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1 Dependencies, functions and data

1.1 Dependencies

```
### Packages
library(emmeans)
library(ggstance)
library(tidyverse)
library(patchwork)
library(data.table)
library(kableExtra)
library(pals)
library(here)
library(lme4)
library(lmerTest)
library(ggh4x)
library(GGally)
library(broom.mixed)
library(survey)
library(lavaan)
library(lavaan.survey)

#### Regarding lavaan.survey
# lavaan.survey is an extension of lavaan that can deal with data independencies.
# It is no longer on cran, but it is a trustworthy package (there is also
# a publication for it). The source code is here
# https://github.com/daob/lavaan.survey.

options(dplyr.summarise.inform = FALSE)
```

1.2 Functions

```
## other useful functions
source(file = here("R/1_Functions/OtherFunctions.R"))
## Permutation entropy function
source(file = here("R/1_Functions/PermutationEntropy.R"))
```

1.3 Data

```
# Time series
dd_list <-
  readRDS(here(
    "Data/2_TimeSeriesData/Transformed/DensityBiomassTimeSeries.RDS"
  ))

# Forecast results
load(here(
  "Data/3_ForecastsResults/3_MergedResults/complete.forecasts.RData"
))
load(here(
  "Data/3_ForecastsResults/3_MergedResults/seg.complete.forecasts.RData"
))
```

2 Implementation of the experiment

2.1 Organisms

```
SpeciesTab <- data.frame(
  Species = c(
    "\\textit{Chlamydomonas}\\n\\textit{reinhardtii}",
    "\\textit{Cryptomonas} sp.",
    "\\textit{Cosmarium}\\n\\textit{botrytis}",
    "\\textit{Desmodesmus}\\n\\textit{armatus}",
    "\\textit{Monoraphidium}\\n\\textit{obtusum}",
    "\\textit{Staurastrum}\\n\\textit{gracile}",
    "\\textit{Staurastrum}\\n\\textit{polytrichum}",
    "\\textit{Colpidium}\\n\\textit{striatum}",
    "\\textit{Dexiostoma}\\n\\textit{campylum}",
    "\\textit{Loxocephalus} sp.",
    "\\textit{Tetrahymena}\\n\\textit{thermophila}",
    "\\textit{Coleps} sp.",
    "\\textit{Paramecium}\\n\\textit{caudatum}",
    "\\textit{Stylonychia}\\n\\textit{mytilus}",
    "\\textit{Stylonychia} sp.",
    "\\textit{Euplotes}\\n\\textit{daidaleos}",
    "\\textit{Paramecium}\\n\\textit{bursaria}",
    "\\textit{Didinium}\\n\\textit{nasutum}"
  ),
  FunctionalGrp = c(
    "Edible\\nalgae",
    "Edible\\nalgae",
    "Inedible\\nalgae",
    "Inedible\\nalgae",
    "Inedible\\nalgae",
    "Inedible\\nalgae",
    "Inedible\\nalgae",
    "Inedible\\nalgae",
    "Bacterivores",
  )
)
```

```

"Bacterivores",
"Bacterivores",
"Bacterivores",
"Omnivores",
"Omnivores",
"Omnivores",
"Omnivores",
"Mixotrophs",
"Mixotrophs",
"Predator"
),
Origin = c(
  "SAG\\textsuperscript{a}",
  "SAG",
  "SAG",
  "SAG",
  "SAG",
  "CCAP\\textsuperscript{b}",
  "SAG",
  "Carolina Biological\\nSupply",
  "CCAP",
  "Carolina Biological\\nSupply",
  "CCAP",
  "Pond\\nUniversity Zurich",
  "Carolina Biological\\nSupply",
  "Pond\\nUniversity Zurich",
  "Pond\\nUniversity Zurich",
  "Sciento",
  "SAG",
  "Sciento"
),
StrainNumber = c(
  "11-32b",
  "26.80",
  "136.80",
  "276-4d",
  "55.81",
  "679/3",
  "20.98",
  NA,
  "1611/1",
  NA,
  "1630/1U",
  NA,
  NA,
  NA,
  NA,
  "P229",
  "27.96",
  "P220"
),
Prey = c(
  NA,

```

```

NA,
NA,
NA,
NA,
NA,
NA,
"Bacteria",
"Bacteria",
"Bacteria",
"Bacteria",
"\\textit{Cryptomonas},\\nBacteria",
"\\textit{Chlamydomonas},\\nBacteria",
"\\textit{Chlamydomonas},\\nBacteria",
"\\textit{Chlamydomonas},\\nBacteria",
"\\textit{Cryptomonas},\\nBacteria",
"\\textit{Chlamydomonas},\\nBacteria",
"\\textit{Paramecium}\\n\\textit{caudatum}"
),
indistFrom = c(
  NA,
  NA,
  NA,
  NA,
  NA,
  NA,
  NA,
  NA,
  NA,
  "\\textit{Tetrahymena}",
  NA,
  "\\textit{Dexiostoma}",
  NA,
  NA,
  "\\textit{Stylonychia} sp.",
  "\\textit{Stylonychia mytilus}",
  NA,
  NA,
  NA
)
)

options(knitr.kable.NA = '')
SpeciesTab %>%
  mutate_all(linebreak, align="l") %>%
  kbl(format = "latex",
      col.names = linebreak(c("Species", "Functional\\ngroup",
                              "Origin", "Strain\\nnumber", "Prey", "Indistin-\\nguishable\\nfrom"),
                              align = "c"),
      caption = "Algae and ciliates used in the experiment.",
      escape = F,
      booktabs = T,
      linesep = "",
      label = "SpeciesTab") %>%
  kable_styling(font_size = 7, latex_options = "striped") %>%

```

Table 1: Algae and ciliates used in the experiment.

Species	Functional group	Origin	Strain number	Prey	Indistinguishable from
<i>Chlamydomonas reinhardtii</i>	Edible algae	SAG ^a	11-32b		
<i>Cryptomonas</i> sp.	Edible algae	SAG	26.80		
<i>Cosmarium botrytis</i>	Inedible algae	SAG	136.80		
<i>Desmodesmus armatus</i>	Inedible algae	SAG	276-4d		
<i>Monoraphidium obtusum</i>	Inedible algae	SAG	55.81		
<i>Staurastrum gracile</i>	Inedible algae	CCAP ^b	679/3		
<i>Staurastrum polytrichum</i>	Inedible algae	SAG	20.98		
<i>Colpidium striatum</i>	Bacterivores	Carolina Biological Supply		Bacteria	
<i>Dexiostoma campylum</i>	Bacterivores	CCAP	1611/1	Bacteria	<i>Tetrahymena</i>
<i>Loxocephalus</i> sp.	Bacterivores	Carolina Biological Supply		Bacteria	
<i>Tetrahymena thermophila</i>	Bacterivores	CCAP	1630/1U	Bacteria	<i>Dexiostoma</i>
<i>Coleps</i> sp.	Omnivores	Pond University Zurich		<i>Cryptomonas</i> , Bacteria	
<i>Paramecium caudatum</i>	Omnivores	Carolina Biological Supply		<i>Chlamydomonas</i> , Bacteria	
<i>Stylonychia mytilus</i>	Omnivores	Pond University Zurich		<i>Chlamydomonas</i> , Bacteria	<i>Stylonychia</i> sp.
<i>Stylonychia</i> sp.	Omnivores	Pond University Zurich		<i>Chlamydomonas</i> , Bacteria	<i>Stylonychia mytilus</i>
<i>Euplotes daidaleos</i>	Mixotrophs	Sciento	P229	<i>Cryptomonas</i> , Bacteria	
<i>Paramecium bursaria</i>	Mixotrophs	SAG	27.96	<i>Chlamydomonas</i> , Bacteria	
<i>Didinium nasutum</i>	Predator	Sciento	P220	<i>Paramecium caudatum</i>	

^a SAG: Culture Collection of Algae at Göttingen University^b CCAP: Culture Collection of Algae and Protozoa

```
footnote(alphabet = c("SAG: Culture Collection of Algae at Göttingen University",
                      "CCAP: Culture Collection of Algae and Protozoa"))
```

2.2 Experimental design

```
# plot the planned change of stressor levels over the course of the experiment
stressor_levels <-
  read.csv(here("Data/1_ExperimentDesignInfo/LightDeclineSchedule.csv"))

# change to long format
stressor_levels_long <-
  gather(stressor_levels, level, value = value, constant:declining)
```

```

stressor_levels_long %>%
  mutate(level = ifelse(level == "declining", "Decreasing", "Constant")) %>%
  ggplot(aes(day, value, col = level)) +
  scale_color_manual(values = c("#C4961A", "#293352"),
                     aesthetics = c("color", "fill")) +
  geom_point() +
  labs(x = "Sampling day", y = "Light level (%)", col = "Light conditions") +
  theme_bw() +
  theme(legend.position = "bottom")

```

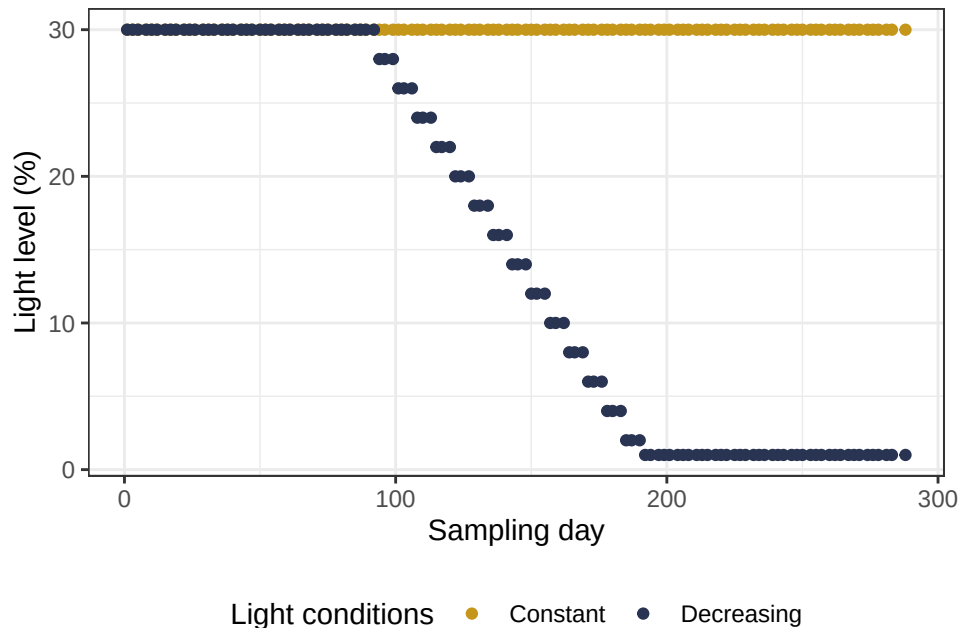


Figure 1: Light levels over the course of the experiment. After 3 months of constant environmental conditions, bottles either remained at the initial light level (30% light) or experienced a gradual decline in light for three months. Then, bottles remained three more months at the final light levels (30 or 1% light intensity). Each point represents a sampling day.

```

PlannedRichTab <- data.frame(
  Tot = c(7, 10, 14),
  Edible = c(2, 2, 2),
  Inedible = c(1, 2, 3),
  Bacterivores = c(1, 2, 3),
  Omni = c(1, 2, 3),
  Mixo = c(1, 2, 2),
  Pred = 1
)

PlannedRichTab %>%
  kbl(
    format = "latex",
    col.names = linebreak(
      c(

```

Table 2: Species richness per functional group in the four diversity levels. Numbers in brackets are the number of available species.

Total (18)	Edible algae (2)	Inedible algae (5)	Bacterivores (4)	Omnivores (4)	Mixotrophs (2)	Predator (1)
7	2	1	1	1	1	1
10	2	2	2	2	2	1
14	2	3	3	3	2	1

```

    "Total\n(18)",
    "Edible algae\n(2)",
    "Inedible algae\n(5)",
    "Bacterivores\n(4)",
    "Omnivores\n(4)",
    "Mixotrophs\n(2)",
    "Predator\n(1)"
  ),
  align = "c"
),
caption = "Species richness per functional group in the four diversity
levels. Numbers in brackets are the number of available species. ",
escape = F,
align = "c",
booktabs = T,
linesep = "",
label = "PlannedRichTab"
)

```

```

design <-
  read.csv(here("Data/1_ExperimentDesignInfo/ExperimentalDesign.csv")) %>%
  mutate(lightTreatment = ifelse(light_treatment == "constant",
                                "Constant", "Decreasing")) %>%
  dplyr::select(bottle, composition, richness, lightTreatment, incubator)

design %>%
  kbl(
    booktabs = T,
    col.names = c(
      "Bottle",
      "Composition",
      "Planned richness",
      "Light conditions",
      "Incubator"
    ),
    caption = "The summarized experimental design.",
    # escape = F,
    align = "lcclc",
    linesep = "",
    label = "designTab"
  ) %>%
  kable_styling(latex_options = "striped", font_size = 8)

```

Table 3: The summarized experimental design.

Bottle	Composition	Planned richness	Light conditions	Incubator
b_01	c_08	10	Constant	A
b_02	c_02	7	Constant	A
b_03	c_11	14	Constant	A
b_04	c_05	7	Constant	A
b_05	c_13	14	Decreasing	B
b_06	c_11	14	Decreasing	B
b_07	c_06	10	Decreasing	B
b_08	c_01	7	Decreasing	B
b_09	c_02	7	Decreasing	C
b_10	c_07	10	Decreasing	C
b_11	c_08	10	Decreasing	C
b_12	c_14	14	Decreasing	C
b_13	c_06	10	Constant	D
b_14	c_03	7	Constant	D
b_15	c_14	14	Constant	D
b_16	c_15	14	Constant	D
b_17	c_05	7	Decreasing	E
b_18	c_04	7	Decreasing	E
b_19	c_12	14	Decreasing	E
b_20	c_09	10	Decreasing	E
b_21	c_07	10	Constant	F
b_22	c_10	10	Constant	F
b_23	c_04	7	Constant	F
b_24	c_13	14	Constant	F
b_25	c_03	7	Decreasing	G
b_26	c_15	14	Decreasing	G
b_27	c_10	10	Decreasing	G
b_28	c_09	10	Constant	H
b_29	c_12	14	Constant	H
b_30	c_01	7	Constant	H

2.3 Selection of compositions

```
comps <-  
  read.csv(here("Data/1_ExperimentDesignInfo/Compositions.csv"))  
rich <- comps$richness  
  
comps2 <- comps %>%  
  dplyr::select(-richness) %>%  
  t() %>%  
  as.data.frame()  
  
colnames(comps2) <- gsub("_", "\\_\\_", comps2[2, 1:15])  
  
comps2 <- comps2 %>%  
  slice_tail(n = 18) %>%  
  mutate_if(is.character, as.numeric)  
  
comps2 <- cbind(Species = rownames(comps2), comps2)  
rownames(comps2) <- NULL  
  
comps2 <- rbind(comps2, c("Planned richness", rich))  
  
comps2 <- comps2 %>%  
  mutate(  
    Species = case_when(  
      Species == "Chlamydomonas_reinhardtii" ~ "\\textit{Chlamydomonas}\\n\\textit{reinhardtii}",  
      Species == "Cryptomonas_sp" ~ "\\textit{Cryptomonas} sp.\\n ",  
      Species == "Monoraphidium_obtusum" ~ "\\textit{Monoraphidium}\\n\\textit{obtusum}",  
      Species == "Cosmarium_botrytis" ~ "\\textit{Cosmarium}\\n\\textit{botrytis}",  
      Species == "Staurastrum_gracile" ~ "\\textit{Staurastrum}\\n\\textit{gracile}",  
      Species == "Staurastrum_polytrichum" ~ "\\textit{Staurastrum}\\n\\textit{polytrichum}",  
      Species == "Desmodesmus_armatus" ~ "\\textit{Desmodesmus}\\n\\textit{armatus}",  
      Species == "Tetrahymena_thermophila" ~ "\\textit{Tetrahymena}\\n\\textit{thermophila}",  
      Species == "Colpidium_striatum" ~ "\\textit{Colpidium}\\n\\textit{striatum}",  
      Species == "Loxocephalus_sp" ~ "\\textit{Loxocephalus} sp.\\n ",  
      Species == "Dexiostoma_campylum" ~ "\\textit{Dexiostoma}\\n\\textit{campylum}",  
      Species == "Paramecium_caudatum" ~ "\\textit{Paramecium}\\n\\textit{caudatum}",  
      Species == "Stylonychia_mytilus" ~ "\\textit{Stylonychia}\\n\\textit{mytilus}",  
      Species == "Stylonychia_sp" ~ "\\textit{Stylonychia} sp.\\n ",  
      Species == "Coleps_sp" ~ "\\textit{Coleps} sp.\\n ",  
      Species == "Paramecium_bursaria" ~ "\\textit{Paramecium}\\n\\textit{bursaria}",  
      Species == "Euplotes_daidaleos" ~ "\\textit{Euplotes}\\n\\textit{daidaleos}",  
      Species == "Didinium_nasutum" ~ "\\textit{Didinium}\\n\\textit{nasutum}",  
      Species == "Planned richness" ~ "Planned\\nrichness"  
    )  
  )  
  
comps2 %>%  
  mutate_all(linebreak, align = "l") %>%  
  kbl(  
    booktabs = T,  
    caption = "The 15 community compositions used in the experiment.  
    A 1 indicates that a species was used in a specific composition,
```

Table 4: The 15 community compositions used in the experiment. A 1 indicates that a species was used in a specific composition, while a 0 indicates that it was not.

Species	Community composition														
	c_01	c_02	c_03	c_04	c_05	c_06	c_07	c_08	c_09	c_10	c_11	c_12	c_13	c_14	c_15
<i>Chlamydomonas reinhardtii</i>	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
<i>Cryptomonas</i> sp.	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
<i>Monoraphidium obtusum</i>	0	0	1	0	0	1	0	0	1	0	0	1	1	0	1
<i>Cosmarium botrytis</i>	0	0	0	1	0	0	0	1	0	1	1	1	1	0	0
<i>Staurastrum gracile</i>	0	1	0	0	0	0	1	0	1	0	1	1	0	1	0
<i>Staurastrum polytrichum</i>	1	0	0	0	0	0	1	0	0	1	0	0	1	1	1
<i>Desmodesmus armatus</i>	0	0	0	0	1	1	0	1	0	0	1	0	0	1	1
<i>Tetrahymena thermophila</i>	1	1	0	0	0	0	0	1	1	0	1	0	1	0	0
<i>Colpidium striatum</i>	0	0	0	0	1	0	1	0	1	1	1	1	1	1	1
<i>Loxocephalus</i> sp.	0	0	1	0	0	1	1	1	0	0	1	1	1	1	1
<i>Dexiostoma campylum</i>	0	0	0	1	0	1	0	0	0	1	0	1	0	1	1
<i>Paramecium caudatum</i>	0	1	0	0	0	1	1	1	0	0	1	1	1	1	1
<i>Stylonychia mytilus</i>	1	0	0	1	0	1	0	0	1	0	0	0	1	1	0
<i>Stylonychia</i> sp.	0	0	1	0	0	0	1	0	0	1	1	1	0	0	1
<i>Coleps</i> sp.	0	0	0	0	1	0	0	1	1	1	1	1	1	1	1
<i>Paramecium bursaria</i>	1	0	0	0	1	0	1	1	1	0	1	1	1	1	1
<i>Euplotes daidaleos</i>	0	1	1	1	0	1	0	0	0	1	1	1	1	1	1
<i>Didinium nasutum</i>	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Planned richness	7	7	7	7	7	10	10	10	10	10	14	14	14	14	14

```

while a 0 indicates that it was not.",
escape = F,
align = c("l", rep("c", 15)),
linesep = "",
label = "CompsTab"
) %>%
kable_styling(latex_options = "striped", font_size = 7) %>%
row_spec(19, bold = T) %>%
add_header_above(header = c(" " = 1, "Community composition" = 15)) %>%
column_spec(2:15, width = "4mm") %>%
row_spec(18, hline_after = T)

```

3 Sampling of the experiment

3.1 Algae

```
flowcam.selection.wide <-  
  read.csv(here("Data/1_ExperimentDesignInfo/FlowCamFiltering.csv"))  
  
flowcam.selection.wide %>%  
  mutate(species = case_when(species == "Chlamydomonas" ~ "Chlamydomonas reinhardtii",  
                             species == "ChlamydomonasClumps" ~ "Chlamydomonas clumps",  
                             species == "Cosmarium" ~ "Cosmarium botrytis",  
                             species == "Debris" ~ "Debris",  
                             species == "Desmodesmus" ~ "Desmodesmus armatus",  
                             species == "DesmodesmusClumps" ~ "Digested algae",  
                             species == "DigestedAlgae" ~ "Digested algae",  
                             species == "DividingChlamydomonas" ~ "Dividing Chlamydomonas",  
                             species == "Monoraphidium" ~ "Monoraphidium obtusum",  
                             species == "Staurostrum1" ~ "Staurostrum gracile",  
                             species == "Staurostrum2" ~ "Staurostrum polytrichum")) %>%  
  ggplot(aes(unfiltered, filtered, col = species)) +  
  geom_point(alpha = 0.7) +  
  geom_abline(intercept = log10(1), slope = 1) +  
  labs(x = expression(paste(  
    "Density in unfiltered sample (cells ", mL ^ {  
      -1  
    }, ")"  
  )),  
       y = expression(paste(  
    "Density in filtered sample (cells ", mL ^ {  
      -1  
    }, ")"  
  )),  
       col = "Taxon") +  
  theme_bw() +  
  scale_color_manual(values = unname(kelly()[2:12])) +  
  scale_x_log10() +  
  scale_y_log10()
```

3.2 Bacteria

```
bacteria.wide <-  
  read.csv(here("Data/1_ExperimentDesignInfo/FlowcytometerFiltering.csv"))  
  
bacteria.wide %>%  
  ggplot(aes(unfiltered, filtered)) +  
  geom_point(alpha = 0.7) +  
  geom_abline(intercept = log10(1), slope = 1) +  
  labs(x = expression(paste(  
    "Density in unfiltered sample (cells ", mL ^ {  
      -1
```

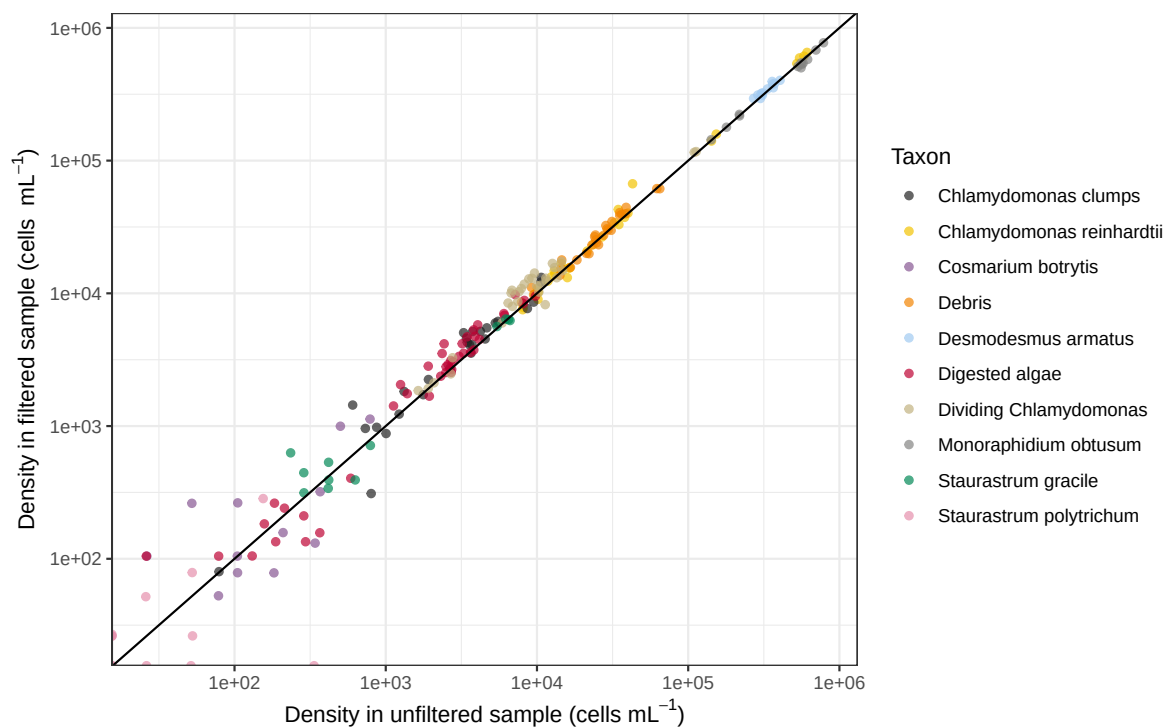


Figure 2: Comparison of filtered and unfiltered algae samples. On one sampling day, we made two flowcam measurements for each of the 30 samples, one from an unfiltered sample and one from a sample that had been filtered through a 200 μ m mesh. The black line is the identity line.

```

    }, ")")
  )),
  y = expression(paste(
    "Density in filtered sample (cells ", mL ^ {
      -1
    }, ")")
  ))) +
  theme_bw() +
  scale_x_log10() +
  scale_y_log10()

```

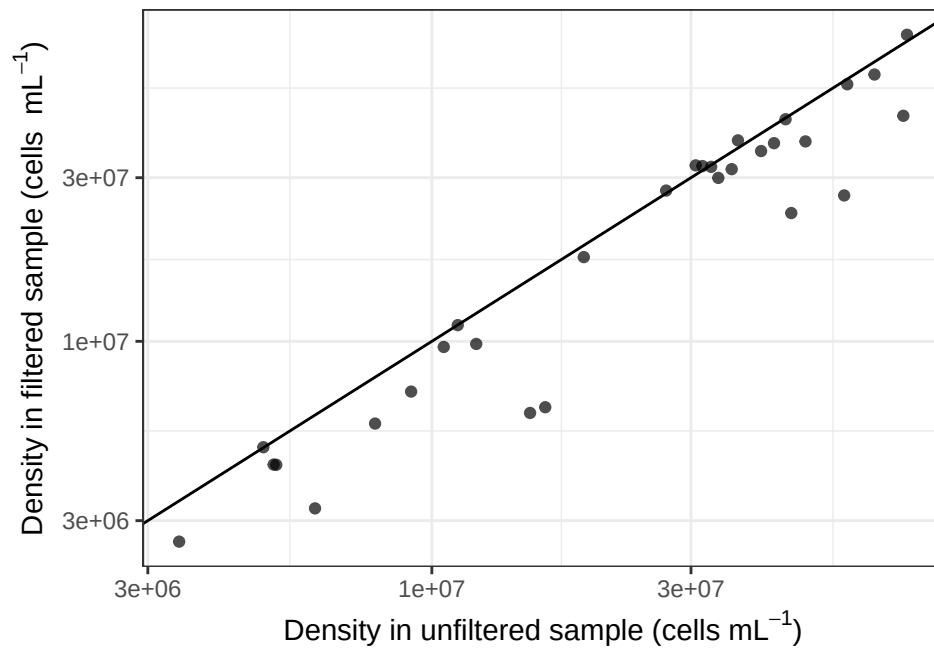


Figure 3: Comparison of filtered and unfiltered bacteria samples. On one sampling day, we made two flowcytometer measurements for each of the 30 samples: one from an unfiltered sample and one from a sample that had been filtered through 200 μm . The black line is the identity line.

4 Quality controls and time series processing

4.1 Quality controls and missing data percentages

```

tab1.1 <- dd_list$densities %>%
  dplyr::filter(method != "Manual count") %>%
  mutate(bottle = gsub("b_", "Bottle ", bottle)) %>%
  group_by(method, bottle) %>%
  summarize(n_NAs = sum(is.na(density)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  dplyr::select(-n_NAs) %>%
  pivot_wider(names_from = method, values_from = perc_NAs) %>%

```

```

  rename(Grouping = bottle)

tab1.2 <- dd_list$densities %>%
  dplyr::filter(method != "Manual count") %>%
  group_by(method, light_treatment) %>%
  dplyr::mutate(
    light_treatment = ifelse(
      light_treatment == "constant",
      "Constant light",
      "Decreasing light"
    )
  ) %>%
  summarize(n_NAs = sum(is.na(density)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  dplyr::select(-n_NAs) %>%
  pivot_wider(names_from = method, values_from = perc_NAs) %>%
  rename(Grouping = light_treatment)

tab1.3 <- dd_list$densities %>%
  dplyr::filter(method != "Manual count") %>%
  group_by(method, richness) %>%
  dplyr::mutate(richness = ifelse(
    richness == 7,
    "Planned richness: 07",
    ifelse(richness == 10, "Planned richness: 10", "Planned richness: 14")
  )) %>%
  summarize(n_NAs = sum(is.na(density)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  dplyr::select(-n_NAs) %>%
  pivot_wider(names_from = method, values_from = perc_NAs) %>%
  rename(Grouping = richness) %>%
  arrange(Grouping)

tab1.4 <- dd_list$densities %>%
  dplyr::filter(method != "Manual count") %>%
  mutate(Grouping = "Experiment") %>%
  group_by(Grouping, method) %>%
  summarize(n_NAs = sum(is.na(density)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  dplyr::select(-n_NAs) %>%
  pivot_wider(names_from = method, values_from = perc_NAs)

tab1 <- rbind(tab1.1, tab1.2, tab1.3, tab1.4)

####

tab2.1 <- dd_list$o2 %>%
  mutate(bottle = gsub("b_", "Bottle ", bottle)) %>%
  group_by(bottle) %>%
  summarize(n_NAs = sum(is.na(percent_o2)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  dplyr::select(-n_NAs) %>%
  rename(Grouping = bottle, Oxygen = perc_NAs)

```

```

tab2.2 <- dd_list$o2 %>%
  group_by(light_treatment) %>%
  dplyr::mutate(
    light_treatment = ifelse(
      light_treatment == "constant",
      "Constant light",
      "Decreasing light"
    )
  ) %>%
  summarize(n_NAs = sum(is.na(percent_o2)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  dplyr::select(-n_NAs) %>%
  rename(Grouping = light_treatment, Oxygen = perc_NAs)

tab2.3 <- dd_list$o2 %>%
  group_by(richness) %>%
  dplyr::mutate(richness = ifelse(
    richness == 7,
    "Planned richness: 07",
    ifelse(richness == 10, "Planned richness: 10", "Planned richness: 14")
  )) %>%
  summarize(n_NAs = sum(is.na(percent_o2)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  dplyr::select(-n_NAs) %>%
  rename(Grouping = richness, Oxygen = perc_NAs) %>%
  arrange(Grouping)

tab2.4 <- dd_list$o2 %>%
  mutate(Grouping = "Experiment") %>%
  group_by(Grouping) %>%
  summarize(n_NAs = sum(is.na(percent_o2)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  rename(Oxygen = perc_NAs) %>%
  dplyr::select(-n_NAs)

tab2 <- rbind(tab2.1, tab2.2, tab2.3, tab2.4)

####

tab3.1 <- dd_list$water_chem %>%
  mutate(bottle = gsub("b_", "Bottle ", bottle)) %>%
  group_by(bottle) %>%
  summarize(n_NAs = sum(is.na(concentration)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  dplyr::select(-n_NAs) %>%
  rename(Grouping = bottle, Water_chemistry = perc_NAs)

tab3.2 <- dd_list$water_chem %>%
  group_by(light_treatment) %>%
  dplyr::mutate(
    light_treatment = ifelse(
      light_treatment == "constant",

```

```

      "Constant light",
      "Decreasing light"
    )
  ) %>%
  summarize(n_NAs = sum(is.na(concentration)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  dplyr::select(-n_NAs) %>%
  rename(Grouping = light_treatment, Water_chemistry = perc_NAs)

tab3.3 <- dd_list$water_chem %>%
  group_by(richness) %>%
  dplyr::mutate(richness = ifelse(
    richness == 7,
    "Planned richness: 07",
    ifelse(richness == 10, "Planned richness: 10", "Planned richness: 14")
  )) %>%
  summarize(n_NAs = sum(is.na(concentration)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  dplyr::select(-n_NAs) %>%
  rename(Grouping = richness, Water_chemistry = perc_NAs) %>%
  arrange(Grouping)

tab3.4 <- dd_list$water_chem %>%
  mutate(Grouping = "Experiment") %>%
  group_by(Grouping) %>%
  summarize(n_NAs = sum(is.na(concentration)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  rename(Water_chemistry = perc_NAs) %>%
  dplyr::select(-n_NAs)

tab3 <- rbind(tab3.1, tab3.2, tab3.3, tab3.4)

###

tab <- full_join(full_join(tab1, tab2), tab3)

# kableExtra::kable(tab)

tab %>%
  # mutate_all(linebreak, align="l") %>%
  kbl(
    "latex",
    booktabs = T,
    col.names = c(
      "Grouping",
      "FlowCam",
      "Flowcytometer",
      "Video",
      "Oxygen",
      "TOC/TN analyzer"
    ),
    caption = "Percentage of missing data for different
    sampling methods and grouping variables.",

```

```

    escape = F,
    align = "lrrrrr",
    linesep = "",
    label = "NAtab"
  ) %>%
  kable_styling(latex_options = "striped", font_size = 8) %>%
  row_spec(36, bold = T) %>%
  add_header_above(header = c(
    " " = 1,
    "Percentage of missing data for different sampling methods" = 5
  )) %>%
  # column_spec(2:5, width = "4mm") %>%
  row_spec(30, hline_after = T) %>%
  row_spec(32, hline_after = T) %>%
  row_spec(35, hline_after = T)

```

4.2 The recorded time series and their processing

```

stripColdd <- dd_list$densities %>%
  ungroup() %>%
  dplyr::select(bottle, light_treatment) %>%
  distinct() %>%
  mutate(stripCol = ifelse(light_treatment == "constant", "#f5d682", "#bdc5de"))

# Only colour strips in x-direction
strip <-
  strip_themed(background_x = elem_list_rect(fill = stripColdd$stripCol))

dd_list$densities %>%
  dplyr::filter(!(
    species %in% c(
      "Algae",
      "Debris",
      "Didinium nasutum",
      "Digested algae",
      "Small unidentified",
      "Colpidium vacuoles"
    )
  ),
  ForecastTarget == T) %>%
  mutate(bottle = gsub("b_", "Bottle ", bottle)) %>%
  ggplot(aes(day, log10(density), col = species)) +
  geom_line(linewidth = 0.6) +
  facet_wrap2( ~ bottle, ncol = 4, strip = strip) +
  theme_bw() +
  theme(
    legend.position = c(0.75, 0.055),
    panel.grid = element_blank(),
    legend.text = element_text(size = 12),
    axis.text = element_text(size = 11),
    axis.title = element_text(size = 16),

```

Table 5: Percentage of missing data for different sampling methods and grouping variables.

Grouping	Percentage of missing data for different sampling methods				
	FlowCam	Flowcytometer	Video	Oxygen	TOC/TN analyzer
Bottle 01	0.81	4.84	0.81	2.42	0.81
Bottle 02	0.81	4.84	0.81	1.61	0.81
Bottle 03	0.81	4.84	0.81	1.61	0.81
Bottle 04	0.81	4.84	0.81	1.61	0.81
Bottle 05	0.81	4.84	0.81	1.61	0.81
Bottle 06	0.81	4.84	0.81	1.61	0.81
Bottle 07	0.81	4.84	0.81	1.61	1.21
Bottle 08	0.81	4.84	0.81	1.61	0.81
Bottle 09	0.81	4.84	0.81	1.61	1.01
Bottle 10	0.81	4.84	0.81	1.61	1.01
Bottle 11	0.81	4.84	0.81	1.61	1.01
Bottle 12	0.81	4.84	0.81	1.61	1.01
Bottle 13	0.81	4.84	0.81	1.61	1.01
Bottle 14	0.81	4.84	0.81	1.61	1.41
Bottle 15	0.81	4.84	0.81	1.61	0.81
Bottle 16	0.81	4.84	0.81	1.61	0.81
Bottle 17	0.81	4.84	0.81	1.61	1.21
Bottle 18	0.81	4.84	1.21	1.61	0.81
Bottle 19	0.81	4.84	0.81	1.61	0.81
Bottle 20	0.81	4.84	0.81	1.61	0.81
Bottle 21	0.81	4.84	0.81	1.61	0.81
Bottle 22	0.81	4.84	0.81	1.61	0.81
Bottle 23	0.81	4.84	1.61	1.61	0.81
Bottle 24	0.81	4.84	0.81	1.61	0.81
Bottle 25	0.81	4.84	1.21	1.61	0.81
Bottle 26	0.81	4.84	0.81	1.61	0.81
Bottle 27	0.81	4.84	0.81	1.61	1.21
Bottle 28	0.81	4.84	0.81	1.61	1.01
Bottle 29	0.81	4.84	0.81	2.42	1.01
Bottle 30	0.81	4.84	0.81	1.61	1.01
Constant light	0.81	4.84	0.83	1.72	0.90
Decreasing light	0.81	4.84	0.85	1.61	0.94
Planned richness: 07	0.81	4.84	0.97	1.61	0.95
Planned richness: 10	0.81	4.84	0.81	1.69	0.97
Planned richness: 14	0.81	4.84	0.81	1.69	0.85
Experiment	0.81	4.84	0.84	1.67	0.92

```

strip.text = element_text(size = 12),
# legend.key.size = unit(3, "line"),
# strip.text = element_text(colour = "white"),
) +
scale_color_manual(values = unname(kelly()[2:16])) +
labs(col = "", y = expression(paste("log"[10], "(Density) (cells ", mL ^
{
-1
}, ")")),
x = "Sampling day") +
guides(col = guide_legend(ncol = 3, byrow = TRUE))

```

```

stripColdd2 <- dd_list$densities %>%
  dplyr::filter(!(
    species %in% c(
      "Algae",
      "Debris",
      "Didinium nasutum",
      "Digested algae",
      "Small unidentified",
      "Colpidium vacuoles"
    )
  ),
  ForecastTarget == F) %>%
  ungroup() %>%
  dplyr::select(bottle, light_treatment) %>%
  distinct() %>%
  mutate(stripCol = ifelse(light_treatment == "constant", "#f5d682", "#bdc5de"))

# Only colour strips in x-direction
strip2 <-
  strip_themed(background_x = elem_list_rect(fill = stripColdd2$stripCol))

dd_list$densities %>%
  dplyr::filter(!(
    species %in% c(
      "Algae",
      "Debris",
      "Didinium nasutum",
      "Digested algae",
      "Small unidentified",
      "Colpidium vacuoles"
    )
  ),
  ForecastTarget == F) %>%
  mutate(bottle = gsub("b_", "Bottle ", bottle)) %>%
  ggplot(aes(day, log10(density), col = species)) +
  geom_line(linewidth = 0.6) +
  facet_wrap2( ~ bottle, ncol = 4, strip = strip2) +
  theme_bw() +
  theme(
    legend.position = c(0.75, 0.07),
    panel.grid = element_blank(),

```

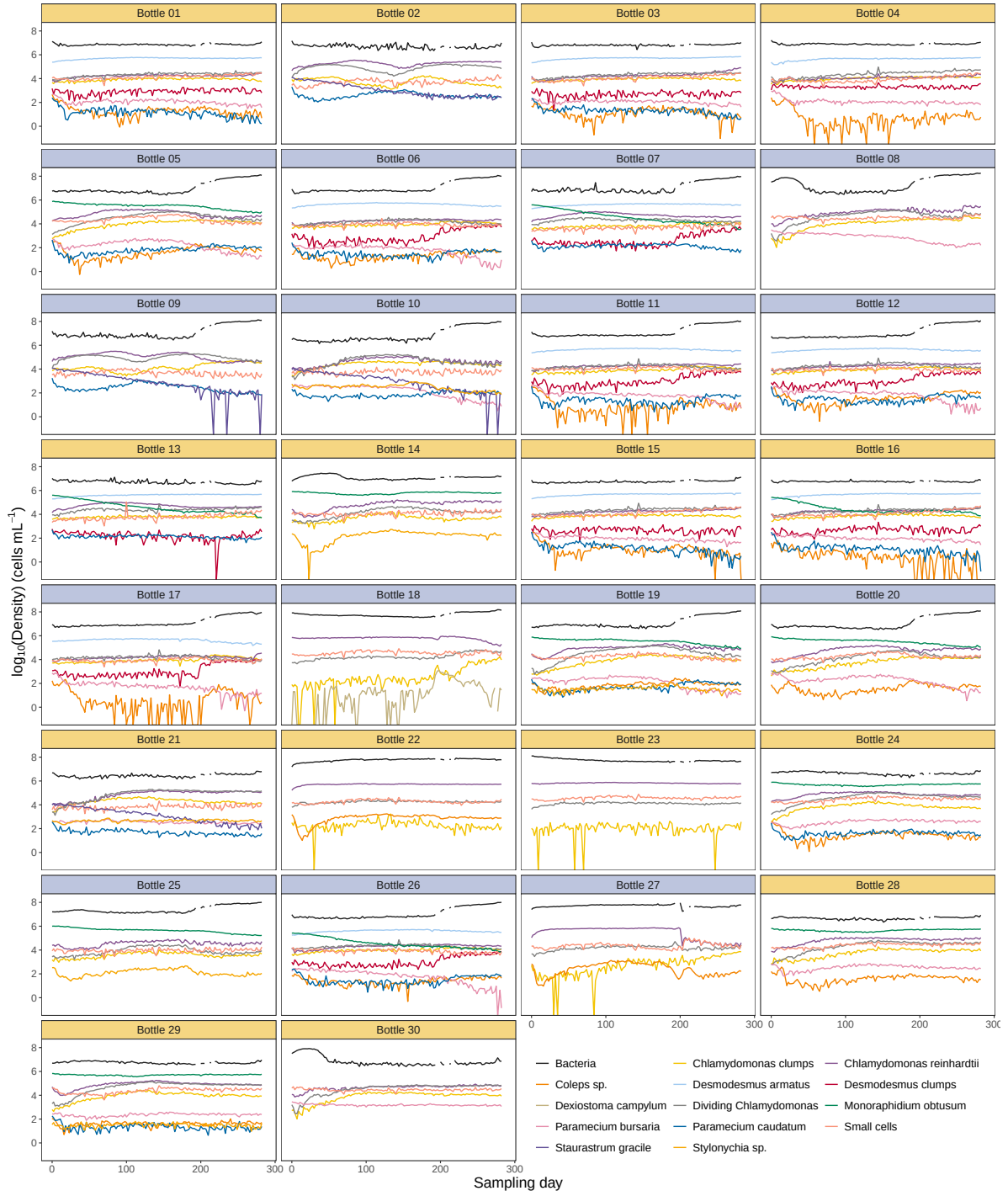


Figure 4: The *untransformed* biotic density time series that were used as forecast targets, shown in the different bottles (i.e. the panels). The time series are coloured by forecast target. The background colour of the subpanel strips indicates whether the bottles were in a constant (orange) or in a decreasing (blue) light environment.

```

legend.text = element_text(size = 12),
axis.text = element_text(size = 11),
axis.title = element_text(size = 16),
strip.text = element_text(size = 12),
# legend.key.size = unit(3, "line"),
# strip.text = element_text(colour = "white"),
) +
scale_color_manual(values = unname(kelly()[2:16])) +
labs(col = "", y = expression(paste("log"[10], "(Density) (cells ", mL ^
{
-1
}, ")")),
x = "Sampling day") +
guides(col = guide_legend(ncol = 3, byrow = TRUE))

```

```

dd_list$biomass %>%
mutate(
  bottle = gsub("b_", "Bottle ", bottle),
  composition = gsub("c_", "Composition ", composition)
) %>%
ggplot(aes(day, log10(biomass), col = composition)) +
geom_line(linewidth = 0.6) +
facet_wrap( ~ light_treatment) +
# facet_wrap2(~bottle, ncol = 4, strip = strip2)+
theme_bw() +
theme(
  legend.position = "bottom",
  panel.grid = element_blank(),
  legend.text = element_text(size = 12),
  axis.text = element_text(size = 11),
  axis.title = element_text(size = 16),
  strip.text = element_text(size = 12),
  # legend.key.size = unit(3, "line"),
  # strip.text = element_text(colour = "white"),
) +
scale_color_manual(values = unname(kelly()[2:16])) +
labs(col = "",
y = expression(paste("log"[10], "(Total biomass) (", g ~ mL ^ {
-1
}, ")")),
x = "Sampling day")

```

```

o2_both <-
read_csv(here("Data/2_TimeSeriesData/Transformed/OxygenBothSensors.csv"))
o2_both <-
full_join(
  o2_both %>% mutate(
    sensor = ifelse(sensor == 4, "Sensor at 400 mL height",
"Sensor at 900 mL height")
),
dd_list$o2 %>% mutate(sensor = "Average of the two sensors")
)

```

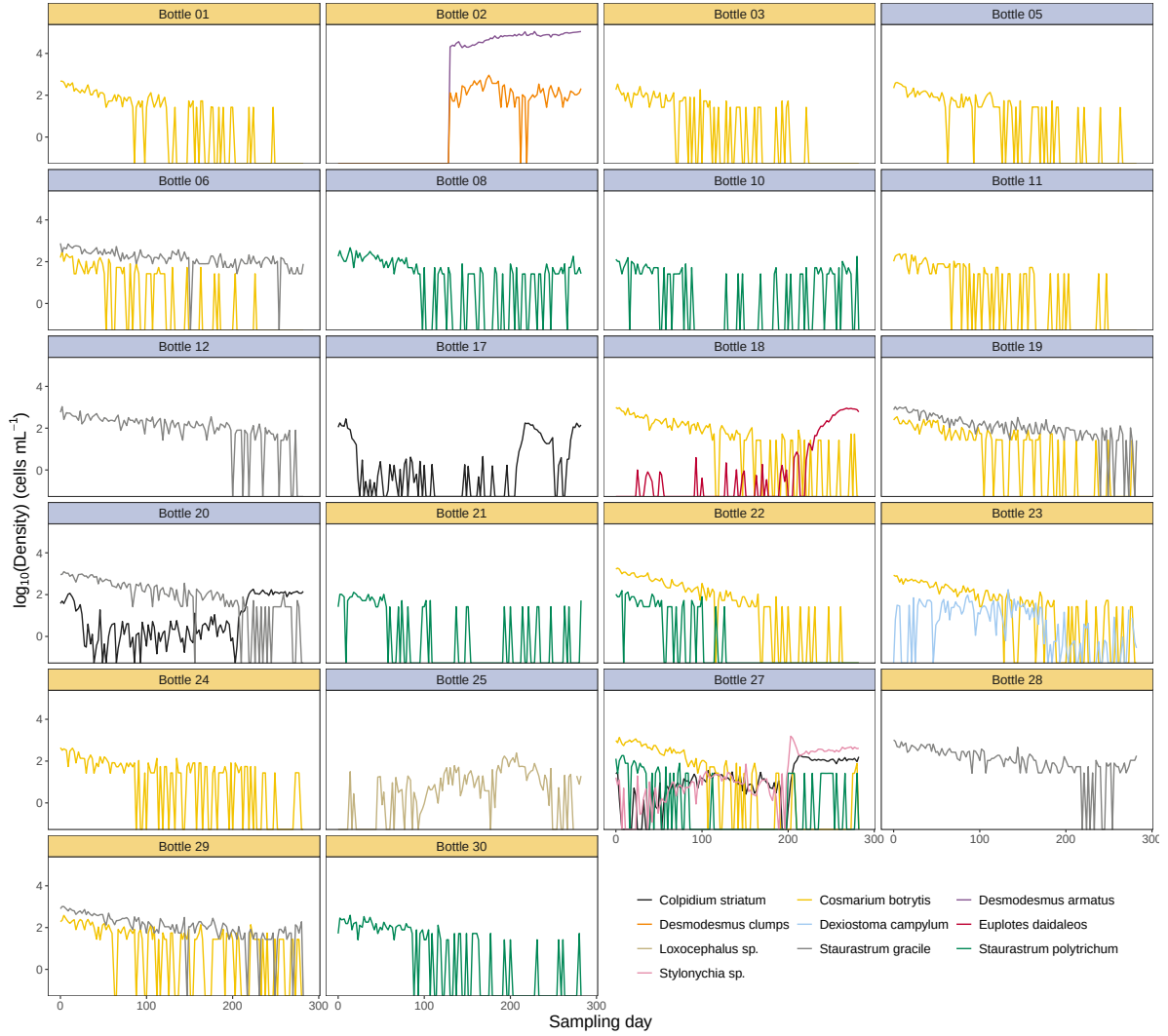


Figure 5: The *untransformed* biotic density time series that were *not* used as forecast targets, shown in the different bottles (i.e. the panels). The time series are coloured by species or species clump. The background colour of the subpanel strips indicates whether the bottles were in a constant (orange) or in a decreasing (blue) light environment.

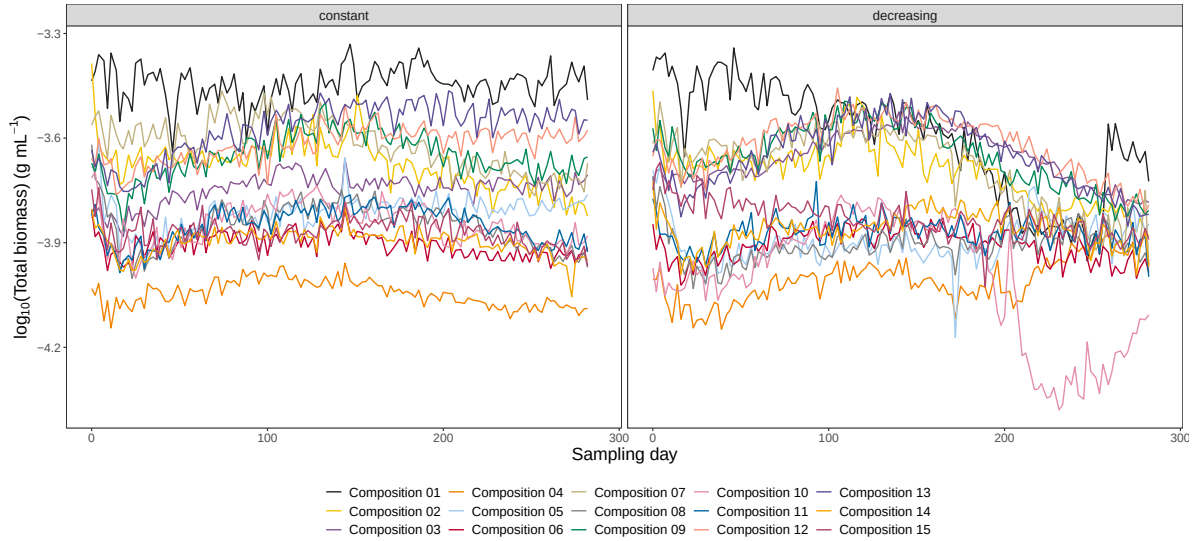


Figure 6: The *untransformed* total biomass time series of the communities, shown in the different light conditions (i.e. the panels). The time series are coloured by community composition.

```
o2_both %>%
  mutate(
    bottle = gsub("b_", "Bottle ", bottle),
    composition = gsub("c_", "Composition ", composition)
  ) %>%
  ggplot(aes(day, percent_o2, col = sensor)) +
  geom_line(linewidth = 0.6) +
  facet_wrap2( ~ bottle,
    ncol = 4,
    strip = strip,
    scales = "free_y") +
  theme_bw() +
  theme(
    legend.position = c(0.75, 0.055),
    panel.grid = element_blank(),
    legend.text = element_text(size = 12),
    axis.text = element_text(size = 11),
    axis.title = element_text(size = 16),
    strip.text = element_text(size = 12),
  ) +
  scale_color_brewer(palette = "Dark2") +
  labs(col = "",
    y = "Oxygen concentration (%)",
    x = "Sampling day")
```

```
dd_list$water_chem %>%
  mutate(type = case_when(
    type == "IC" ~ "IC: Inorganic Carbon",
    type == "TC" ~ "DC: Dissolved Carbon",
    type == "TN" ~ "DN: Dissolved Nitrogen",
    type == "TOC" ~ "DOC: Dissolved Organic Carbon",
  ))
```

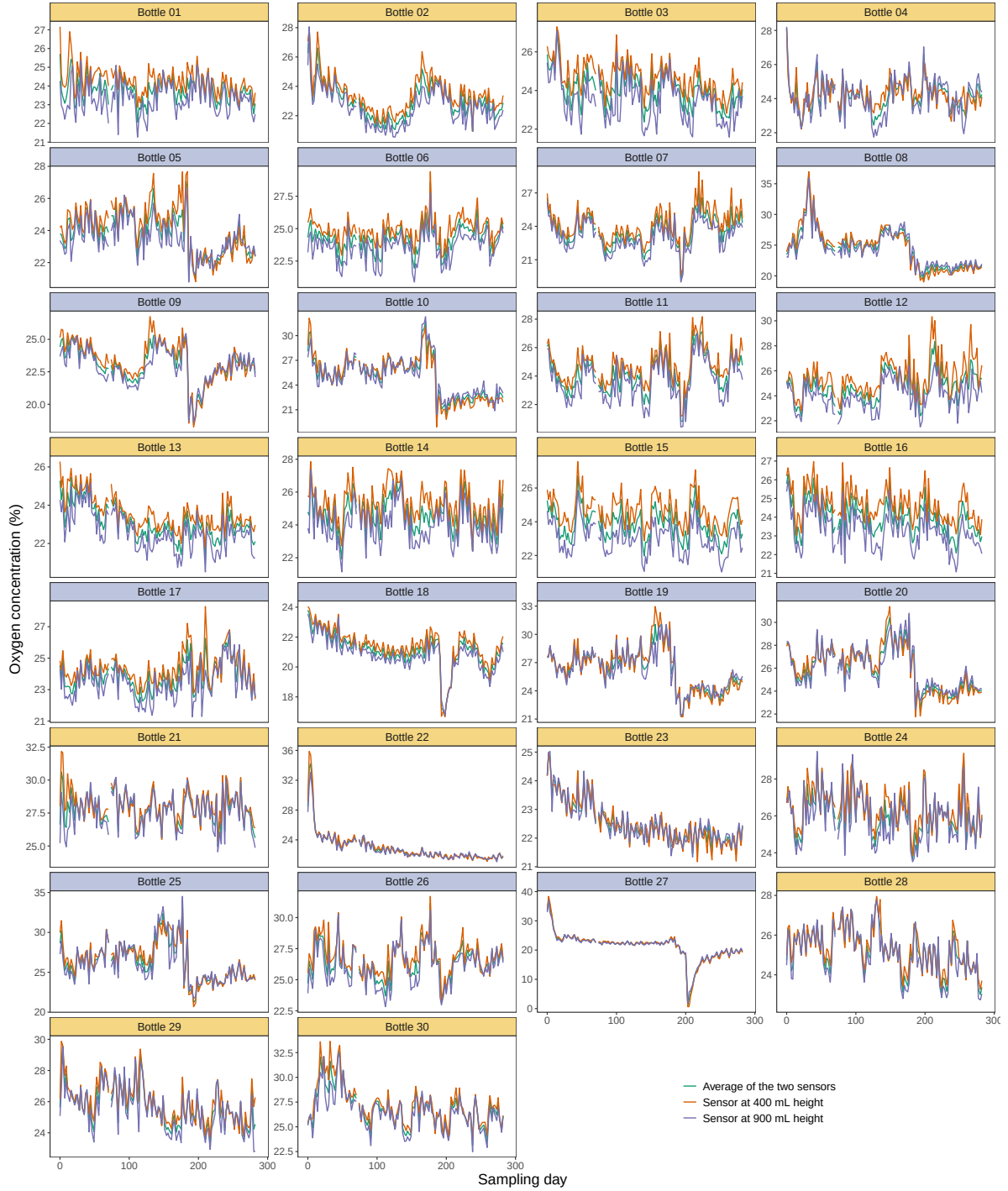


Figure 7: The *untransformed* oxygen concentration time series, shown in the different bottles (i.e. the panels). The time series are coloured by sensor height (with the average of the two time series also shown). The background colour of the subpanel strips indicates whether the bottles were in a constant (orange) or in a decreasing (blue) light environment.

```

    bottle = gsub("b_", "Bottle ", bottle),
    composition = gsub("c_", "Composition ", composition)) %>%
ggplot(aes(day, concentration, col=type)) +
geom_line(linewidth=0.6) +
facet_wrap2(~bottle, ncol = 4, strip = strip, scales = "free_y") +
theme_bw() +
theme(legend.position = c(0.75, 0.055),
      panel.grid=element_blank(),
      legend.text = element_text(size=12),
      axis.text = element_text(size=11),
      axis.title = element_text(size=16),
      strip.text = element_text(size=12),
      ) +
scale_color_brewer(palette = "Dark2") +
labs(col="",
      y = expression(paste("Concentration (", mg~L^{-1}, ")")),
      x = "Sampling day")

```

```

dd_list$densities.int.detrend.complete %>%
dplyr::filter(!(species %in% c("Algae", "Debris", "Didinium nasutum",
                              "Digested algae", "Small unidentified",
                              "Colpidium vacuoles")),
              ForecastTarget==T) %>%
mutate(bottle = gsub("b_", "Bottle ", bottle)) %>%
ggplot(aes(day, density, col=species)) +
geom_line(linewidth=0.6) +
facet_wrap2(~bottle, ncol = 4, strip = strip) +
theme_bw() +
theme(legend.position = c(0.75, 0.055),
      panel.grid=element_blank(),
      legend.text = element_text(size=12),
      axis.text = element_text(size=11),
      axis.title = element_text(size=16),
      strip.text = element_text(size=12),
      # legend.key.size = unit(3, "line"),
      # strip.text = element_text(colour = "white"),
      ) +
scale_color_manual(values = unname(kelly()[2:16])) +
labs(col="", y = "Standardized density",
      x = "Sampling day") +
guides(col=guide_legend(ncol=3, byrow=TRUE))

```

```

dd_list$biomass.int.detrend.complete %>%
mutate(bottle = gsub("b_", "Bottle ", bottle),
      composition = gsub("c_", "Composition ", composition)) %>%
ggplot(aes(day, biomass, col=composition)) +
geom_line(linewidth=0.6) +
facet_wrap(~light_treatment) +
# facet_wrap2(~bottle, ncol = 4, strip = strip2) +
theme_bw() +
theme(legend.position = "bottom",
      panel.grid=element_blank(),

```



Figure 8: The *untransformed* carbon and nitrogen concentration time series, shown in the different bottles (i.e. the panels). The time series are coloured by the type of carbon and nitrogen concentration. The background colour of the subpanel strips indicates whether the bottles were in a constant (orange) or in a decreasing (blue) light environment.

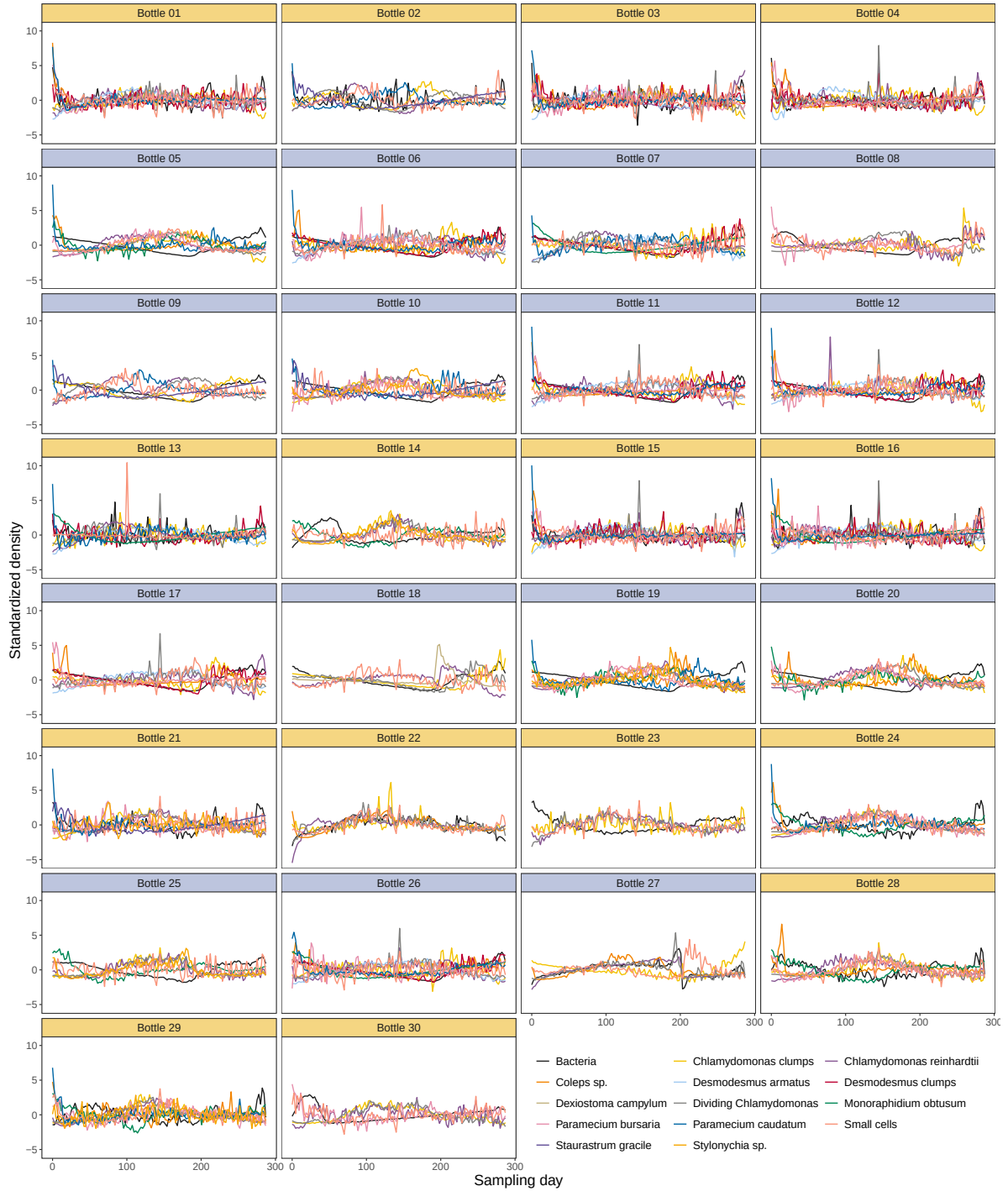


Figure 9: The *transformed* biotic density time series that were used as forecast targets, shown in the different bottles (i.e. the panels). The time series are coloured by forecast target. The background colour of the subpanel strips indicates whether the bottles were in a constant (orange) or in a decreasing (blue) light environment.

```

    legend.text = element_text(size=12),
    axis.text = element_text(size=11),
    axis.title = element_text(size=16),
    strip.text = element_text(size=12),
    # legend.key.size = unit(3,"line"),
    # strip.text = element_text(colour = "white"),
  ) +
  scale_color_manual(values = unname(kelly()[2:16])) +
  labs(col="",
    y = "Standardized total biomass",
    x = "Sampling day")

```

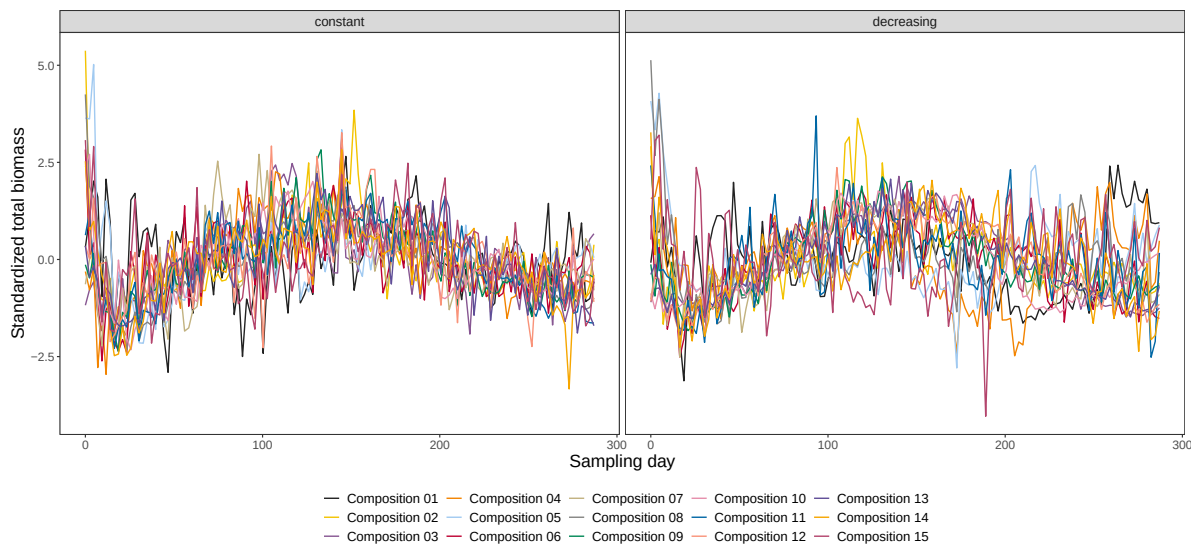


Figure 10: The *transformed* total biomass time series of the communities, shown in the different light conditions (i.e. the panels). The time series are coloured by community composition.

```

dd_list$o2.int.detrend.complete %>%
  mutate(bottle = gsub("b_", "Bottle ", bottle),
    composition = gsub("c_", "Composition ", composition)) %>%
  ggplot(aes(day, percent_o2, col=composition)) +
  geom_line(linewidth=0.6) +
  facet_wrap(~light_treatment) +
  # facet_wrap2(~bottle, ncol = 4, strip = strip2) +
  theme_bw() +
  theme(legend.position = "bottom",
    panel.grid=element_blank(),
    legend.text = element_text(size=12),
    axis.text = element_text(size=11),
    axis.title = element_text(size=16),
    strip.text = element_text(size=12),
    # legend.key.size = unit(3,"line"),
    # strip.text = element_text(colour = "white"),
  ) +
  scale_color_manual(values = unname(kelly()[2:16])) +
  labs(col="",

```

```
y = "Standardized Oxygen concentration",
x = "Sampling day")
```

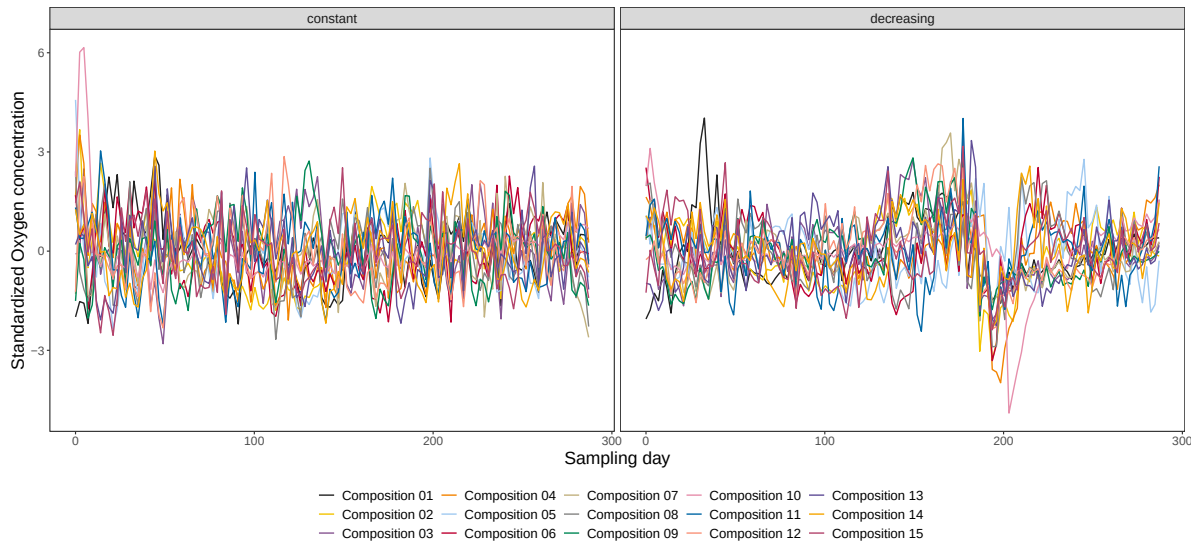


Figure 11: The *transformed* oxygen concentration time series, shown in the different light conditions (i.e. the panels). The time series are coloured by community composition.

5 Calculation of realized richness, biomass and time series properties

5.1 The realized species richness

```
dd.rich.test <- dd_list$densities %>%
  dplyr::filter(day == 287) %>%
  group_by(
    bottle,
    light_treatment,
    mean_richness,
    median_richness,
    composition,
    realized_richness
  ) %>%
  dplyr::summarise(n = n())

t.test.light <-
  t.test(
    mean_richness ~ light_treatment,
    paired = TRUE,
    alternative = "two.sided",
    data = dd.rich.test
  )
t.test.light
```

```
##
## Paired t-test
##
## data: mean_richness by light_treatment
## t = -0.62589, df = 14, p-value = 0.5415
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## -1.2020660 0.6589766
## sample estimates:
## mean difference
## -0.2715447
```

```
lowerfun <- function(data, mapping,...){
  ggplot(data = data, mapping = mapping)+
    geom_abline(slope = 1, intercept = 0)+
    geom_smooth(method = "lm", ...) +
    geom_point(...) +
    theme_bw()
}

my_dens <- function(data, mapping, ...) {
  ggplot(data = data, mapping = mapping) +
    geom_density(..., alpha=0.5) +
    theme_bw()
}

my_hist <- function(data, mapping, ...) {
  ggplot(data = data, mapping = mapping) +
    geom_histogram(..., alpha=0.3, position="identity") +
    theme_bw()
}

## realized species richness

numb.species <- dd_list$densities %>%
  dplyr::filter(
    species %in% c(
      "Bacteria",
      "Chlamydomonas reinhardtii",
      "Coleps sp.",
      "Cosmarium botrytis",
      "Desmodesmus armatus",
      "Paramecium bursaria",
      "Paramecium caudatum",
      "Staurostrum gracile",
      "Monoraphidium obtusum",
      "Staurostrum polytrichum",
      "Stylonychia sp.",
      "Colpidium striatum",
      "Dexiostoma campylum",
      "Euplotes daidaleos",
      "Loxoccephallus sp."
    )
  ) %>%
```

```

group_by(bottle, richness, composition, light_treatment, ForecastTarget) %>%
dplyr::summarize(
  realized_richness = length(unique(species)),
  # persisting.species = list(unique(species)),
  mean_richness = unique(mean_richness),
  median_richness = unique(median_richness)
)

colnames(numb.species) <-
c(
  "bottle",
  "Planned richness",
  "composition",
  "light_treatment",
  "ForecastTarget",
  "Minimal richness",
  "Mean richness",
  "Median richness"
)

cor.numb.species <- numb.species %>%
ggpairs(
  mapping = aes(col = light_treatment, fill = light_treatment),
  columns = c(
    "Planned richness",
    "Minimal richness",
    "Mean richness",
    "Median richness"
  ),
  lower = list(continuous = lowerfun, alpha = 0.5),
  diag = list(continuous = my_hist),
  legend = 1
) +
scale_color_manual(values = c("#C4961A", "#293352"),
  aesthetics = c("color", "fill"))

cor.numb.species +
  labs(col = "Light conditions", fill = "Light conditions") +
  theme(legend.position = "bottom")

dd.rich.test <- dd_list$densities %>%
dplyr::filter(day==287) %>%
group_by(bottle, light_treatment, mean_richness, median_richness, composition,
  realized_richness) %>%
dplyr::summarise(n=n()) %>%
mutate(bottle = gsub("b_", "Bottle ", bottle),
  composition = gsub("c_", "Composition ", composition))

rich.test.1 <- dd.rich.test %>%
ggplot(aes(light_treatment, mean_richness, group=composition))+
geom_boxplot(aes(col=light_treatment, group=light_treatment), show.legend = F)+
geom_line(col="black", alpha=0.3,
  position=position_dodge(width = 0.3))+

```

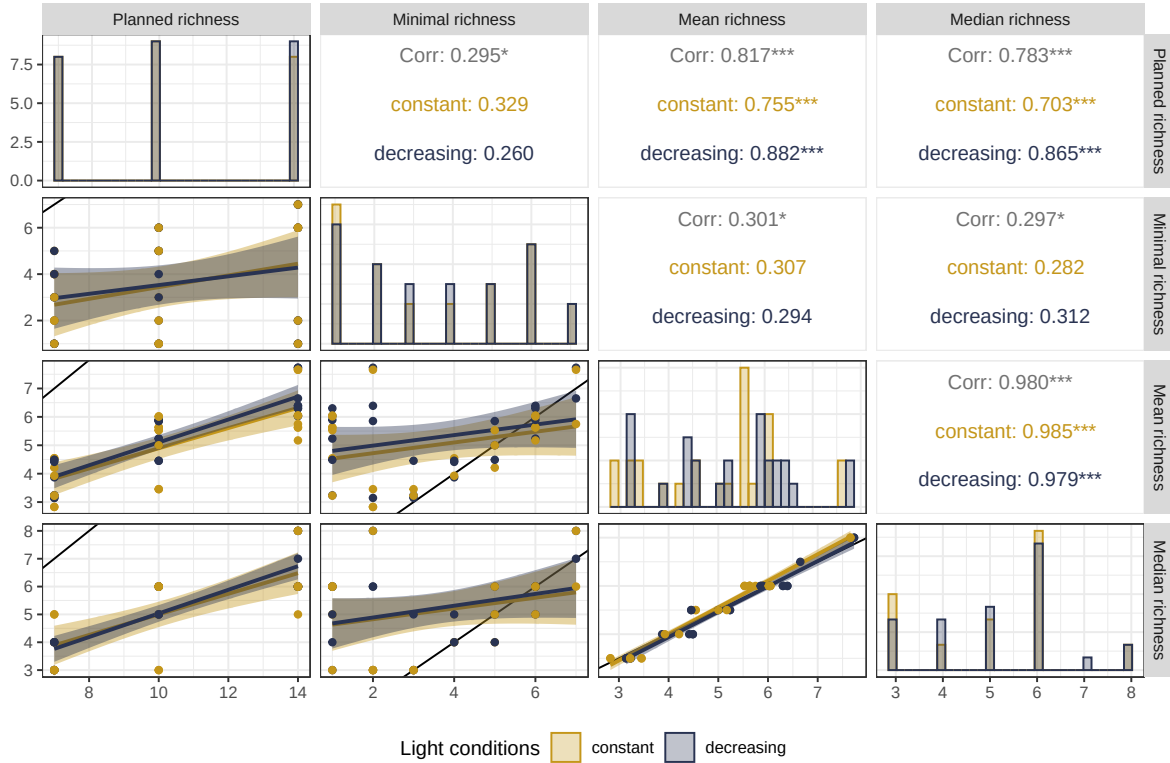


Figure 12: Comparison of the different species richness metrics (i.e. the planned and the realized median, mean and minimal richness), colour-coded for the light conditions. The panels in the diagonal show the bar-plots of the richness variables. The panels below the diagonal show the pairwise scatter-plots as indicated by the facet labels (the black line is the identity line, the coloured lines and the corresponding shaded areas show the fit and the 95% CI of a linear regression). The panels above the diagonal showing the corresponding correlation estimates.

```

geom_point(aes(fill=composition), position=position_dodge(width = 0.3), size=3,
           na.rm = T, alpha=1, shape=21)+
theme_bw()+
theme(legend.position = "bottom")+
scale_fill_manual(values = unname(kelly()[2:16]))+
scale_y_continuous(limits = c(0,8)) +
scale_color_manual(values = c("#C4961A", "#293352"))+
labs(x="Light conditions", y="Mean richness",
     fill="") +
guides(col=guide_legend(ncol = 1,byrow=TRUE))

rich.test.2 <- dd.rich.test %>%
ggplot(aes(light_treatment, median_richness, group=composition))+
geom_boxplot(aes(col=light_treatment, group=light_treatment), show.legend = F)+
geom_line(col="black", alpha=0.3,
          position=position_dodge(width = 0.3))+
geom_point(aes(fill=composition), position=position_dodge(width = 0.3), size=3,
           na.rm = T, alpha=1, shape=21)+
theme_bw()+
theme(legend.position = "bottom")+
scale_fill_manual(values = unname(kelly()[2:16]))+
scale_y_continuous(limits = c(0,8)) +
scale_color_manual(values = c("#C4961A", "#293352"))+
labs(x="Light conditions", y="Median richness",
     fill="") +
guides(col=guide_legend(ncol = 1,byrow=TRUE))

rich.test.3 <- dd.rich.test %>%
ggplot(aes(light_treatment, realized_richness, group=composition))+
geom_boxplot(aes(col=light_treatment, group=light_treatment), show.legend = F)+
geom_line(col="black", alpha=0.3,
          position=position_dodge(width = 0.3))+
geom_point(aes(fill=composition), position=position_dodge(width = 0.3), size=3,
           na.rm = T, alpha=1, shape=21)+
theme_bw()+
theme(legend.position = "bottom")+
scale_fill_manual(values = unname(kelly()[2:16]))+
scale_y_continuous(limits = c(0,8)) +
scale_color_manual(values = c("#C4961A", "#293352"))+
labs(x="Light conditions", y="Minimal richness",
     fill="") +
guides(col=guide_legend(ncol = 1,byrow=TRUE))

rich.test.1 + rich.test.2 + rich.test.3 + plot_layout(guides = "collect") +
plot_annotation(tag_levels = "a") &
theme(legend.position = 'bottom')

```

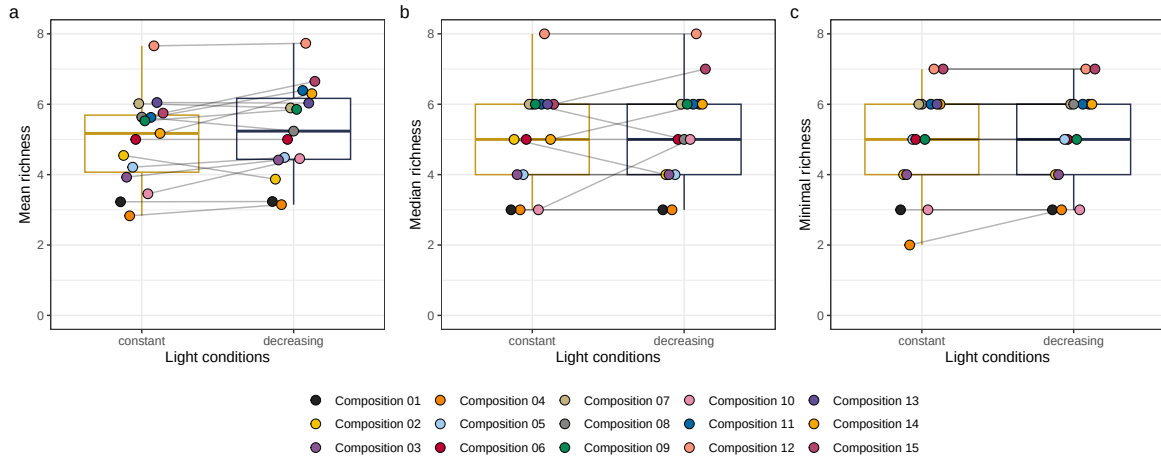


Figure 13: Scatter- and box-plot to investigate the effects of the light conditions on the realized species richness measures. (a) Mean richness; (b) Median richness; (c) Minimal richness. The points are coloured based on species composition and bottles with the same starting species composition are connected by a line. The boxplots are coloured based on light condition (orange = constant, blue = decreasing).

6 Forecasting

6.1 Correlation of forecasts across methods

```
complete.forecasts.wide <- complete.forecasts %>%
  dplyr::select(
    bottle,
    richness,
    composition,
    incubator,
    realized_richness,
    light_treatment,
    Target,
    RMSE,
    forecast.method,
    mean_richness,
    median_richness
  ) %>%
  pivot_wider(names_from = forecast.method, values_from = RMSE)

cor.plot.complete <- complete.forecasts.wide %>%
  ggpairs(
    mapping = aes(col = light_treatment, fill = light_treatment),
    columns = c("ARIMA", "Multiview EDM", "RNN", "Random Forest", "Simplex EDM"),
    lower = list(continuous = lowerfun, alpha = 0.5),
    diag = list(continuous = my_dens),
    legend = 1
  ) +
  scale_color_manual(values = c("#C4961A", "#293352"),
    aesthetics = c("color", "fill"))
```

```
cor.plot.complete +
  labs(col = "Light conditions", fill = "Light conditions") +
  theme(legend.position = "bottom")
```

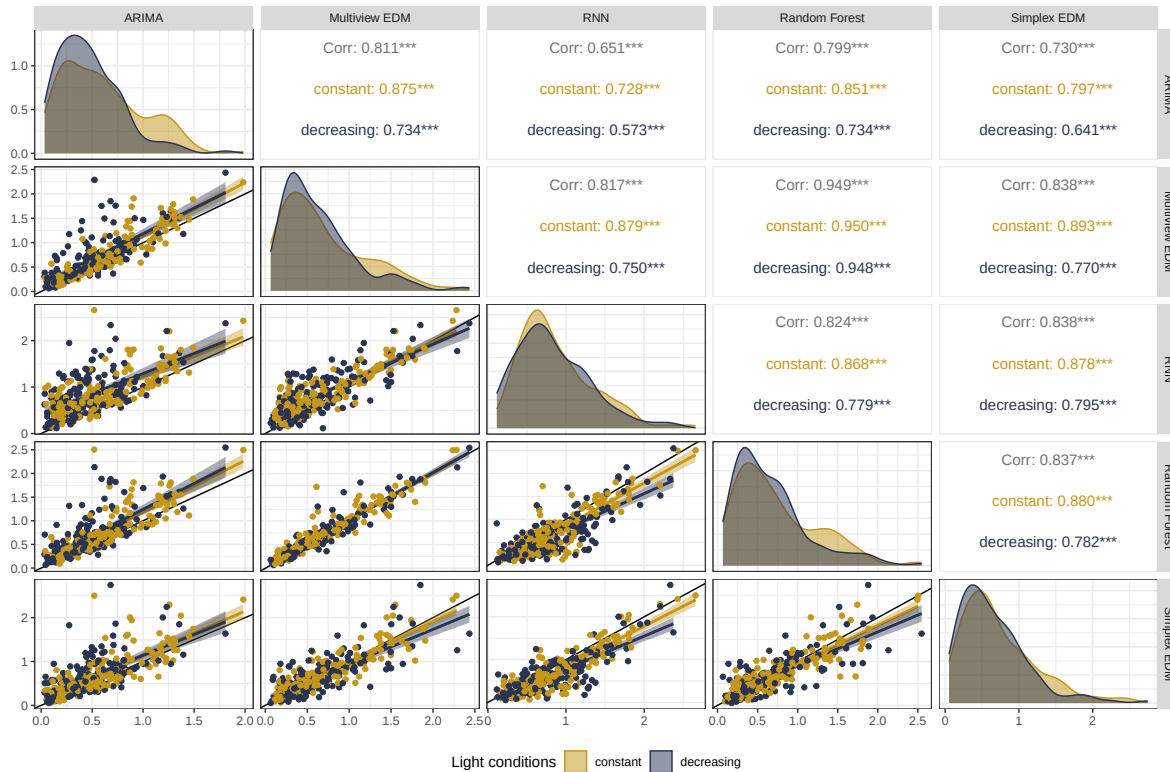


Figure 14: Comparison of forecast errors values (i.e. RMSE) produced by the different forecast methods (i.e. ARIMA, Multiview EDM, RNN, Random Forest and Simplex EDM), colour-coded for the light conditions. The panels in the diagonal show the density plots of the forecast errors. The panels below the diagonal show the pairwise scatter-plots as indicated by the facet labels (the black line is the identity line, the coloured lines and the corresponding shaded areas show the fit and the 95% CI of a linear regression). The panels above the diagonal showing the corresponding correlation estimates.

7 Additional figures and tables

7.1 Effects of experimental conditions on forecasts

```
##### -----
##### Species forecast model regressions
##### -----

formula <-
  as.formula(
    RMSE ~ median_richness_center * light_treatment +
```

```

    (1 | Target) + (1 | bottle) + (1 | incubator)
  )

m.simplex <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(
      forecast.method == "Simplex EDM",
      Target != "Total_biomass",
      Target != "Mean_oxygen"
    )
)

m.multiview <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(
      forecast.method == "Multiview EDM",
      Target != "Total_biomass",
      Target != "Mean_oxygen"
    )
)

m.arima <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(
      forecast.method == "ARIMA",
      Target != "Total_biomass",
      Target != "Mean_oxygen"
    )
)

m.rnn <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(
      forecast.method == "RNN",
      Target != "Total_biomass",
      Target != "Mean_oxygen"
    )
)

m.rf <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(
      forecast.method == "Random Forest",
      Target != "Total_biomass",
      Target != "Mean_oxygen"
    )
)

```

```

m.species.list <- list(
  `Simplex EDM` = m.simplex,
  `Multiview EDM` = m.multiview,
  `ARIMA` = m.arima,
  `RNN` = m.rnn,
  `Random Forest` = m.rf
)

##### -----
##### Total biomass forecast model regressions
##### -----

formula <-
  as.formula(RMSE ~ median_richness_center * light_treatment +
    (1 | composition) + (1 | incubator))

m.simplex.bio <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "Simplex EDM",
      Target == "Total_biomass")
)

m.multiview.bio <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "Multiview EDM",
      Target == "Total_biomass")
)

m.arima.bio <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "ARIMA",
      Target == "Total_biomass")
)

m.rnn.bio <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "RNN",
      Target == "Total_biomass")
)

m.rf.bio <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "Random Forest",
      Target == "Total_biomass")
)

m.bio.list <- list(
  `Simplex EDM` = m.simplex.bio,

```

```

`Multiview EDM` = m.multiview.bio,
`ARIMA` = m.arima.bio,
`RNN` = m.rnn.bio,
`Random Forest` = m.rf.bio
)

##### -----
##### Mean oxygen forecast model regressions
##### -----

formula <-
  as.formula(RMSE ~ median_richness_center * light_treatment +
    (1 | composition) + (1 | incubator))

m.simplex.oxy <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "Simplex EDM",
      Target == "Mean_oxygen")
)

m.multiview.oxy <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "Multiview EDM",
      Target == "Mean_oxygen")
)

m.arima.oxy <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "ARIMA",
      Target == "Mean_oxygen")
)

m.rnn.oxy <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "RNN",
      Target == "Mean_oxygen")
)

m.rf.oxy <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "Random Forest",
      Target == "Mean_oxygen")
)

m.oxy.list <- list(
  `Simplex EDM` = m.simplex.oxy,
  `Multiview EDM` = m.multiview.oxy,
  `ARIMA` = m.arima.oxy,

```

```

`RNN` = m.rnn.oxy,
`Random Forest` = m.rf.oxy
)

##### -----
##### Total biomass vs Mean Species forecast model regressions
##### -----

average.complete.forecasts2 <- complete.forecasts %>%
  dplyr::filter(!(Target %in% c("Mean_oxygen")) %>%
    mutate(aggregateMeasure = ifelse(Target %in% c("Total_biomass", "Mean_oxygen"),
                                          T, F)) %>%

  group_by(
    bottle,
    richness,
    composition,
    incubator,
    realized_richness,
    mean_richness,
    median_richness,
    light_treatment,
    forecast.method,
    median_richness_center,
    mean_richness_center,
    aggregateMeasure
  ) %>%
  summarise(meanRMSE = mean(RMSE),
            sdRMSE = sd(RMSE))

formula <-
  as.formula(
    meanRMSE ~ median_richness_center * light_treatment +
      median_richness_center * aggregateMeasure +
      light_treatment * aggregateMeasure +
      (1 | bottle) + (1 | incubator)
  )

m.simplex.aggr.bio <- lmer(
  formula,
  average.complete.forecasts2 %>%
    dplyr::filter(forecast.method == "Simplex EDM")
)

m.multiview.aggr.bio <-
  lmer(
    formula,
    average.complete.forecasts2 %>%
      dplyr::filter(forecast.method == "Multiview EDM")
  )

m.arima.aggr.bio <- lmer(formula,
  average.complete.forecasts2 %>%
    dplyr::filter(forecast.method == "ARIMA"))

```

```

m.rnn.aggr.bio <- lmer(formula,
                      average.complete.forecasts2 %>%
                      dplyr::filter(forecast.method == "RNN"))

m.rf.aggr.bio <- lmer(
  formula,
  average.complete.forecasts2 %>%
  dplyr::filter(forecast.method == "Random Forest")
)

m.aggr.bio.list <- list(
  `Simplex EDM` = m.simplex.aggr.bio,
  `Multiview EDM` = m.multiview.aggr.bio,
  `ARIMA` = m.arima.aggr.bio,
  `RNN` = m.rnn.aggr.bio,
  `Random Forest` = m.rf.aggr.bio
)

##### -----
##### Mean oxygen vs Mean Species forecast model regressions
##### -----

average.complete.forecasts.oxy <- complete.forecasts %>%
  dplyr::filter(!(Target %in% c("Total_biomass"))) %>%
  mutate(aggregatedMeasure = ifelse(Target %in% c("Mean_oxygen", "Total_biomass"),
                                     T, F)) %>%

  group_by(
    bottle,
    richness,
    composition,
    incubator,
    realized_richness,
    mean_richness,
    median_richness,
    light_treatment,
    forecast.method,
    median_richness_center,
    mean_richness_center,
    aggregatedMeasure
  ) %>%
  summarise(meanRMSE = mean(RMSE),
            sdRMSE = sd(RMSE))

formula <-
  as.formula(
    meanRMSE ~ median_richness_center * light_treatment +
      median_richness_center * aggregatedMeasure +
      light_treatment * aggregatedMeasure +
      (1 | bottle) + (1 | incubator)
  )

m.simplex.aggr.oxy <-

```

```

lmer(
  formula,
  average.complete.forecasts.oxy %>%
    dplyr::filter(forecast.method == "Simplex EDM")
)

m.multiview.aggr.oxy <-
  lmer(
    formula,
    average.complete.forecasts.oxy %>%
      dplyr::filter(forecast.method == "Multiview EDM")
  )

m.arima.aggr.oxy <- lmer(
  formula,
  average.complete.forecasts.oxy %>%
    dplyr::filter(forecast.method == "ARIMA")
)

m.rnn.aggr.oxy <- lmer(formula,
  average.complete.forecasts.oxy %>%
    dplyr::filter(forecast.method == "RNN"))

m.rf.aggr.oxy <- lmer(
  formula,
  average.complete.forecasts.oxy %>%
    dplyr::filter(forecast.method == "Random Forest")
)

m.aggr.oxy.list <- list(
  `Simplex EDM` = m.simplex.aggr.oxy,
  `Multiview EDM` = m.multiview.aggr.oxy,
  `ARIMA` = m.arima.aggr.oxy,
  `RNN` = m.rnn.aggr.oxy,
  `Random Forest` = m.rf.aggr.oxy
)

```

```

#### -----
#### Table for simplex species forecasts regression (Main text)
#### -----

```

```

r.simplex <- rand(m.simplex)

spec.ab.rmse.tab <-
  data.frame(
    Response = "Forecast error\nof target\nabundances",
    Covariate = c(
      "Intercept",
      "Richness",
      "Light (decreasing)",
      "Richness:Light",
      "Target (taxa)",
      "Bottle",

```

```

      "Incubator"
    ),
    Type = c(
      rep("Fixed\\neffect", 4),
      "Random\\nintercept",
      "Random\\nintercept",
      "Random\\nintercept"
    ),
    Estimate = c(fixef(m.simplex), as.data.frame(VarCorr(m.simplex))$sdcor[1:3]),
    DF = c(round(summary(m.simplex)$coef[, 3, drop = FALSE], 0), r.simplex$Df[2:4]),
    SE = c(summary(m.simplex)$coef[, 2, drop = FALSE], NA, NA, NA),
    Test = c(rep("\\textit{t}", 4), "$\\chi^2$", "$\\chi^2$", "$\\chi^2$"),
    Test.value = c(summary(m.simplex)$coef[, 4, drop = FALSE], r.simplex$LRT[2:4]),
    p.value = c(summary(m.simplex)$coef[, 5, drop = FALSE],
      r.simplex$Pr(>Chisq)`[2:4])
  )

spec.ab.rmse.tab.anova.tab <-
  data.frame(
    Response = "Forecast error\\nof target\\nabundances",
    Covariate = c("Richness", "Light (decreasing)", "Richness:Light"),
    Type = "Sum of\\nsquares",
    Estimate = anova(m.simplex)[, 1],
    DF = paste(anova(m.simplex)[, 3], round(anova(m.simplex)[, 4], 1), sep = ", "),
    SE = NA,
    Test = "\\textit{F}",
    Test.value = anova(m.simplex)[, 5],
    p.value = anova(m.simplex)[, 6]
  )

spec.ab.rmse.tab <-
  rbind(spec.ab.rmse.tab, spec.ab.rmse.tab.anova.tab)

#### -----
#### Table for species forecasts regression (Supplementary): all forecasts
#### -----

# needed? it's basically the robustness analysis, which includes additional
# information

#### -----
#### Table for simplex community biomass regression (Main text)
#### -----

r.simplex.bio <- rand(m.simplex.bio)

bio.rmse.tab <-
  data.frame(
    Response = "Forecast error\\nof community\\nbiomass",
    Covariate = c(
      "Intercept",

```

```

      "Richness",
      "Light (decreasing)",
      "Richness:Light",
      "Composition",
      "Incubator"
    ),
    Type = c(
      rep("Fixed\\neffect", 4),
      "Random\\nintercept",
      "Random\\nintercept"
    ),
    Estimate = c(fixef(m.simplex.bio), as.data.frame(VarCorr(m.simplex.bio))$sdcor[1:2]),
    DF = c(round(summary(m.simplex.bio)$coef[, 3, drop = FALSE], 0), r.simplex.bio$Df[2:3]),
    SE = c(summary(m.simplex.bio)$coef[, 2, drop = FALSE], NA, NA),
    Test = c(rep("\\textit{t}", 4), "$\\chi^2$", "$\\chi^2$"),
    Test.value = c(summary(m.simplex.bio)$coef[, 4, drop = FALSE], r.simplex.bio$LRT[2:3]),
    p.value = c(summary(m.simplex.bio)$coef[, 5, drop = FALSE],
      r.simplex.bio$`Pr(>Chisq)`[2:3])
  )
)

bio.rmse.anova.tab <-
  data.frame(
    Response = "Forecast error\\nof community\\nbiomass",
    Covariate = c("Richness", "Light (decreasing)", "Richness:Light"),
    Type = "Sum of\\nsquares",
    Estimate = anova(m.simplex.bio)[, 1],
    DF = paste(anova(m.simplex.bio)[, 3], round(anova(m.simplex.bio)[, 4], 1),
      sep = ", "),
    SE = NA,
    Test = "\\textit{F}",
    Test.value = anova(m.simplex.bio)[, 5],
    p.value = anova(m.simplex.bio)[, 6]
  )

bio.rmse.tab <- rbind(bio.rmse.tab, bio.rmse.anova.tab)

#### -----
#### Table for community biomass regression (Supplementary): all forecasts
#### -----

newdat.bio <- data.frame()
tab.bio <- data.frame()

df <- complete.forecasts %>%
  dplyr::filter(Target == "Total_biomass")

newdat.b <-
  expand.grid(
    median_richness_center = seq(
      min(df$median_richness_center),
      max(df$median_richness_center),
      length.out = 100
    )
  )

```

```

    ),
    light_treatment = unique(df$light_treatment)
  )

for (m.name in names(m.bio.list)) {
  m <- m.bio.list[[m.name]]
  newdat.b2 <-
    conf_interval(m, newdat.b, "RMSE", "Forecast error (RMSE)")
  newdat.b2$forecast.method <- m.name

  r <- rand(m)

  tab1 <-
    data.frame(
      Response = "Forecast\nerror of\ncommunity\nbiomass",
      forecast.method = m.name,
      Covariate = c(
        "Intercept",
        "Richness",
        "Light (decreasing)",
        "Richness:Light",
        "Composition",
        "Incubator"
      ),
      Type = c(
        rep("Fixed\neffect", 4),
        "Random\nintercept",
        "Random\nintercept"
      ),
      Estimate = c(fixef(m), as.data.frame(VarCorr(m))$sdcor[1:2]),
      DF = c(round(summary(m)$coef[, 3, drop = FALSE], 0), r$Df[2:3]),
      SE = c(summary(m)$coef[, 2, drop = FALSE], NA, NA),
      Test = c(rep("\textit{t}", 4), "$\\chi^2$", "$\\chi^2$"),
      Test.value = c(summary(m)$coef[, 4, drop = FALSE], r$LRT[2:3]),
      p.value = c(summary(m)$coef[, 5, drop = FALSE], r$`Pr(>Chisq)`[2:3])
    )

  tab2 <-
    data.frame(
      Response = "Forecast\nerror of\ncommunity\nbiomass",
      forecast.method = m.name,
      Covariate = c("Richness", "Light (decreasing)", "Richness:Light"),
      Type = "Sum of\nsquares",
      Estimate = anova(m)[, 1],
      DF = paste(anova(m)[, 3], round(anova(m)[, 4], 1), sep = ", "),
      SE = NA,
      Test = "\textit{F}",
      Test.value = anova(m)[, 5],
      p.value = anova(m)[, 6]
    )

  tab1 <- rbind(tab1, tab2)
  newdat.bio <- rbind(newdat.bio, newdat.b2)
}

```

```

tab.bio <- rbind(tab.bio, tab1)
}

#### -----
#### Table for mean oxygen regression (Supplementary): all forecasts
#### -----

newdat.oxy <- data.frame()
tab.oxy <- data.frame()

df <- complete.forecasts %>%
  dplyr::filter(Target == "Mean_oxygen")

newdat.o <-
  expand.grid(
    median_richness_center = seq(
      min(df$median_richness_center),
      max(df$median_richness_center),
      length.out = 100
    ),
    light_treatment = unique(df$light_treatment)
  )

for (m.name in names(m.oxy.list)) {
  m <- m.oxy.list[[m.name]]
  newdat.o2 <-
    conf_interval(m, newdat.o, "RMSE", "Forecast error (RMSE)")
  newdat.o2$forecast.method <- m.name

  r <- rand(m)

  tab1 <- data.frame(
    Response = "Forecast\nerror\nof mean\noxygen",
    forecast.method = m.name,
    Covariate = c(
      "Intercept",
      "Richness",
      "Light (decreasing)",
      "Richness:Light",
      "Composition",
      "Incubator"
    ),
    Type = c(
      rep("Fixed\neffect", 4),
      "Random\nintercept",
      "Random\nintercept"
    ),
    Estimate = c(fixef(m), as.data.frame(VarCorr(m))$sdcor[1:2]),
    DF = c(round(summary(m)$coef[, 3, drop = FALSE], 0), r$Df[2:3]),
    SE = c(summary(m)$coef[, 2, drop = FALSE], NA, NA),
    Test = c(rep("\\textit{t}", 4), "\\chi^2$", "\\chi^2$"),
    Test.value = c(summary(m)$coef[, 4, drop = FALSE], r$LRT[2:3]),

```

```

    p.value = c(summary(m)$coef[, 5, drop = FALSE], r$`Pr(>Chisq)`[2:3])
  )

  tab2 <- data.frame(
    Response = "Forecast\nerror\nof mean\noxygen",
    forecast.method = m.name,
    Covariate = c("Richness", "Light (decreasing)", "Richness:Light"),
    Type = "Sum of\nsquares",
    Estimate = anova(m)[, 1],
    DF = paste(anova(m)[, 3], round(anova(m)[, 4], 1), sep = ", "),
    SE = NA,
    Test = "\\textit{F}",
    Test.value = anova(m)[, 5],
    p.value = anova(m)[, 6]
  )

  tab1 <- rbind(tab1, tab2)
  newdat.oxy <- rbind(newdat.oxy, newdat.o2)
  tab.oxy <- rbind(tab.oxy, tab1)
}

#### -----
#### Table for aggregate analysis: Biomass vs Species forecasts: all forecasts
#### -----

newdat.aggr.bio <- data.frame()
tab.aggr.bio <- data.frame()
em.tab.bio <- data.frame()

newdat.a.b <- with(
  average.complete.forecasts2,
  expand.grid(
    median_richness_center = seq(
      min(median_richness_center),
      max(median_richness_center),
      length.out = 100
    ),
    light_treatment = unique(light_treatment),
    aggregatedMeasure = unique(aggregatedMeasure)
  )
)

for (m.name in names(m.aggr.bio.list)) {
  m <- m.aggr.bio.list[[m.name]]
  newdat.a.b2 <-
    conf_interval(m, newdat.a.b, "meanRMSE", "Mean forecast error (RMSE)")
  newdat.a.b2$forecast.method <- m.name

  em <- emmeans(m, c("aggregatedMeasure"), by = "light_treatment")
  em <- as.data.frame(pairs(em))
  em$forecast.method = m.name
}

```

```

em.tab.bio <- rbind(em.tab.bio, em)

r <- rand(m)

tab1 <-
  data.frame(
    Response = "Mean forecast\nerror of\nabundances\n \nVersus\n \nForecast error\nof community\nbiom",
    forecast.method = m.name,
    Covariate = c(
      "Intercept",
      "Richness",
      "Light (decreasing)",
      "Measure (biomass)",
      "Richness:Light",
      "Richness:Measure",
      "Light:Measure",
      "Bottle",
      "Incubator"
    ),
    Type = c(
      rep("Fixed\neffect", 7),
      "Random\nintercept",
      "Random\nintercept"
    ),
    Estimate = c(fixef(m), as.data.frame(VarCorr(m))$sdcor[1:2]),
    DF = c(round(summary(m)$coef[, 3, drop = FALSE], 0), r$Df[2:3]),
    SE = c(summary(m)$coef[, 2, drop = FALSE], NA, NA),
    Test = c(rep("\\textit{t}", 7), "\\chi^2$", "\\chi^2$"),
    Test.value = c(summary(m)$coef[, 4, drop = FALSE], r$LRT[2:3]),
    p.value = c(summary(m)$coef[, 5, drop = FALSE], r$`Pr(>Chisq)`[2:3])
  )

tab2 <-
  data.frame(
    Response = "Mean forecast\nerror of\nabundances\n \nVersus\n \nForecast error\nof community\nbiom",
    forecast.method = m.name,
    Covariate = c(
      "Richness",
      "Light (decreasing)",
      "Measure (biomass)",
      "Richness:Light",
      "Richness:Measure",
      "Light:Measure"
    ),
    Type = "Sum of\nsquares",
    Estimate = anova(m)[, 1],
    DF = paste(anova(m)[, 3], round(anova(m)[, 4], 1), sep = ", "),
    SE = NA,
    Test = "\\textit{F}",
    Test.value = anova(m)[, 5],
    p.value = anova(m)[, 6]
  )

```

```

tab1 <- rbind(tab1, tab2)
newdat.aggr.bio <- rbind(newdat.aggr.bio, newdat.a.b2)
tab.aggr.bio <- rbind(tab.aggr.bio, tab1)
}

#### -----
#### Table for aggregate analysis: Oxygen vs Species forecasts: all forecasts
#### -----

newdat.aggr.oxy <- data.frame()
tab.aggr.oxy <- data.frame()
em.tab.oxy <- data.frame()

newdat.ox <- with(
  average.complete.forecasts.oxy,
  expand.grid(
    median_richness_center = seq(
      min(median_richness_center),
      max(median_richness_center),
      length.out = 100
    ),
    light_treatment = unique(light_treatment),
    aggregatedMeasure = unique(aggregatedMeasure)
  )
)

for (m.name in names(m.aggr.oxy.list)) {
  m <- m.aggr.oxy.list[[m.name]]
  newdat.ox2 <-
    conf_interval(m, newdat.ox, "meanRMSE", "Mean forecast error (RMSE)")
  newdat.ox2$forecast.method <- m.name

  em <- emmeans(m, c("aggregatedMeasure"), by = "light_treatment")
  em <- as.data.frame(pairs(em))
  em$forecast.method = m.name
  em.tab.oxy <- rbind(em.tab.oxy, em)

  r <- rand(m)

  tab1 <-
    data.frame(
      Response = "Mean forecast\nerror of\nabundances\n\nVersus\n\nForecast error\nof mean\noxygen\nno",
      forecast.method = m.name,
      Covariate = c(
        "Intercept",
        "Richness",
        "Light (decreasing)",
        "Measure (oxygen)",
        "Richness:Light",
        "Richness:Measure",
        "Light:Measure",
        "Bottle",

```

```

      "Incubator"
    ),
    Type = c(
      rep("Fixed\\neffect", 7),
      "Random\\nintercept",
      "Random\\nintercept"
    ),
    Estimate = c(fixef(m), as.data.frame(VarCorr(m))$sdcor[1:2]),
    DF = c(round(summary(m)$coef[, 3, drop = FALSE], 0), r$Df[2:3]),
    SE = c(summary(m)$coef[, 2, drop = FALSE], NA, NA),
    Test = c(rep("\\textit{t}", 7), "\\chi^2$", "\\chi^2$"),
    Test.value = c(summary(m)$coef[, 4, drop = FALSE], r$LRT[2:3]),
    p.value = c(summary(m)$coef[, 5, drop = FALSE], r$`Pr(>Chisq)`[2:3])
  )
}

tab2 <-
  data.frame(
    Response = "Mean forecast\\nerror of\\nabundances\\n \\nVersus\\n \\nForecast error\\nof mean\\noxygen\\noxygen",
    forecast.method = m.name,
    Covariate = c(
      "Richness",
      "Light (decreasing)",
      "Measure (oxygen)",
      "Richness:Light",
      "Richness:Measure",
      "Light:Measure"
    ),
    Type = "Sum of\\nsquares",
    Estimate = anova(m)[, 1],
    DF = paste(anova(m)[, 3], round(anova(m)[, 4], 1), sep = ", "),
    SE = NA,
    Test = "\\textit{F}",
    Test.value = anova(m)[, 5],
    p.value = anova(m)[, 6]
  )

tab1 <- rbind(tab1, tab2)
newdat.aggr.oxy <- rbind(newdat.aggr.oxy, newdat.ox2)
tab.aggr.oxy <- rbind(tab.aggr.oxy, tab1)
}

#### -----
#### Table for marginal means: aggregated measures vs non-aggregated
#### -----

em.tab.oxy <- em.tab.oxy %>%
  mutate(contrast = "Species - Oxygen") %>%
  dplyr::select(contrast,
    forecast.method,
    light_treatment,
    estimate,

```

```

        SE,
        df,
        t.ratio,
        p.value)

em.tab.bio <- em.tab.bio %>%
  mutate(contrast = "Species - Biomass") %>%
  dplyr::select(contrast,
               forecast.method,
               light_treatment,
               estimate,
               SE,
               df,
               t.ratio,
               p.value)

em.tab <- rbind(em.tab.bio, em.tab.oxy)

main.tab <- rbind(spec.ab.rmse.tab, bio.rmse.tab)

rownames(main.tab) <- NULL

main.tab$p.value <- ifelse(main.tab$p.value < 0.0001, "<0.0001",
                          sprintf("%.4f", round(main.tab$p.value, 4)))

main.tab <- main.tab %>%
  dplyr::mutate(
    Estimate = round(Estimate, 3),
    SE = round(SE, 3),
    Test.value = round(Test.value, 3)
  )

options(knitr.kable.NA = '-')
main.tab %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 3,
    col.names = c(
      "Response",
      "Covariate",
      "Type",
      "Estimate",
      "DF",
      "SE",
      "Test",
      "Value",
      "\\textit{p}-value"
    ),
    caption = "Results of linear mixed models investigating the effects of
species richness and light on forecast errors of taxa and community biomass.

```

Table 6: Results of linear mixed models investigating the effects of species richness and light on forecast errors of taxa and community biomass. This table is an extended version of Table 1 in the main text and includes both the regression tables and the analysis of variance (Type III) tables of the models. The models included median centered richness, light treatment and their interaction as fixed effects. The random intercepts were included for the bottles, incubators and the targets (i.e. taxa). Forecast error is based on forecasts computed with Simplex EDM.

Response	Covariate	Type	Estimate	DF	SE	Test	Value	p-value
Forecast error of target abundances	Intercept	Fixed effect	0.685	16	0.093	<i>t</i>	7.371	<0.0001
	Richness		0.056	22	0.033	<i>t</i>	1.695	0.1044
	Light (decreasing)		-0.095	6	0.086	<i>t</i>	-1.109	0.3092
	Richness:Light		-0.102	20	0.046	<i>t</i>	-2.246	0.0363
	Target (taxa)	Random intercept	0.074	1	-	χ^2	47.396	<0.0001
	Bottle		0.250	1	-	χ^2	0.410	0.5219
	Incubator		0.085	1	-	χ^2	1.198	0.2737
	Richness	Sum of squares	0.008	1, 21.1	-	<i>F</i>	0.046	0.8328
	Light (decreasing)		0.225	1, 6.1	-	<i>F</i>	1.230	0.3092
	Richness:Light		0.922	1, 19.7	-	<i>F</i>	5.046	0.0363
Forecast error of community biomass	Intercept	Fixed effect	0.668	21	0.111	<i>t</i>	6.036	<0.0001
	Richness		0.039	25	0.073	<i>t</i>	0.529	0.6017
	Light (decreasing)		-0.034	13	0.105	<i>t</i>	-0.329	0.7476
	Richness:Light		-0.046	13	0.075	<i>t</i>	-0.622	0.5445
	Composition	Random intercept	0.303	1	-	χ^2	4.702	0.0301
	Incubator		0.000	1	-	χ^2	0.000	1.0000
	Richness	Sum of squares	0.005	1, 19.5	-	<i>F</i>	0.060	0.8089
	Light (decreasing)		0.008	1, 13.1	-	<i>F</i>	0.108	0.7476
	Richness:Light		0.029	1, 13.4	-	<i>F</i>	0.387	0.5445

This table is an extended version of Table 1 in the main text and includes both the regression tables and the analysis of variance (Type III) tables of the models. The models included median centered richness, light treatment and their interaction as fixed effects. The random intercepts were included for the bottles, incubators and the targets (i.e. taxa). Forecast error is based on forecasts computed with Simplex EDM.",

```
align = c("lllrrrcrr"),
escape = F,
label = "MainTab"
) %>%
collapse_rows(
  columns = c(1, 3),
  latex_hline = 'custom',
  custom_latex_hline = c(1, 3),
  valign = "middle"
) %>%
kable_styling(font_size = 7)
```

```
complete.forecasts %>%
  dplyr::filter(Target %in% c("Total_biomass")) %>%
  ggplot(aes(median_richness_center, RMSE, col = light_treatment)) +
  geom_ribbon(
    data = newdat.bio,
    aes(
      ymin = lower,
      ymax = upper,
```

```

    y = fit,
    group = light_treatment
  ),
  col = NA,
  fill = "grey",
  alpha = 0.3
) +
geom_line(data = newdat.bio,
          linewidth = 1.3,
          aes(y = fit, group = light_treatment)) +
geom_point(size = 2.5, alpha = 0.6) +
facet_grid( ~ forecast.method) +
labs(x = "Median realized species richness", y = "Forecast error of\ncommunity biomass (RMSE)",
     col = "Light conditions") +
scale_color_manual(values = c("#C4961A", "#293352"),
                   aesthetics = c("color", "fill")) +
theme_bw() +
theme(legend.position = "bottom")

```

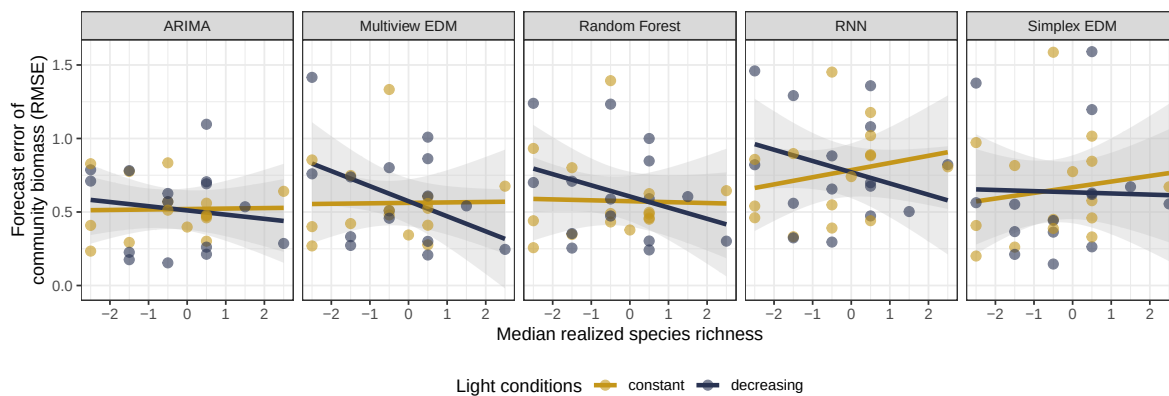


Figure 15: Relation between species richness and light and the forecast error of community biomass. Relationship between forecast error of oxygen and richness in constant light (yellow) and declining light (blue). Lines display the fit of the linear mixed model, shaded areas denote the 95% confidence intervals.

```

complete.forecasts %>%
  dplyr::filter(Target %in% c("Mean_oxygen")) %>%
  ggplot(aes(median_richness_center, RMSE, col = light_treatment)) +
  geom_ribbon(
    data = newdat.oxy,
    aes(
      ymin = lower,
      ymax = upper,
      y = fit,
      group = light_treatment
    ),
    col = NA,
    fill = "grey",
    alpha = 0.3
  ) +

```

```

geom_line(data = newdat.oxy,
          linewidth = 1.3,
          aes(y = fit, group = light_treatment)) +
geom_point(size = 2.5, alpha = 0.6) +
facet_grid( ~ forecast.method) +
labs(x = "Median realized species richness",
     y = "Forecast error of\noxygen concentration (RMSE)",
     col = "Light conditions") +
scale_color_manual(values = c("#C4961A", "#293352"),
                   aesthetics = c("color", "fill")) +
theme_bw() +
theme(legend.position = "bottom")

```

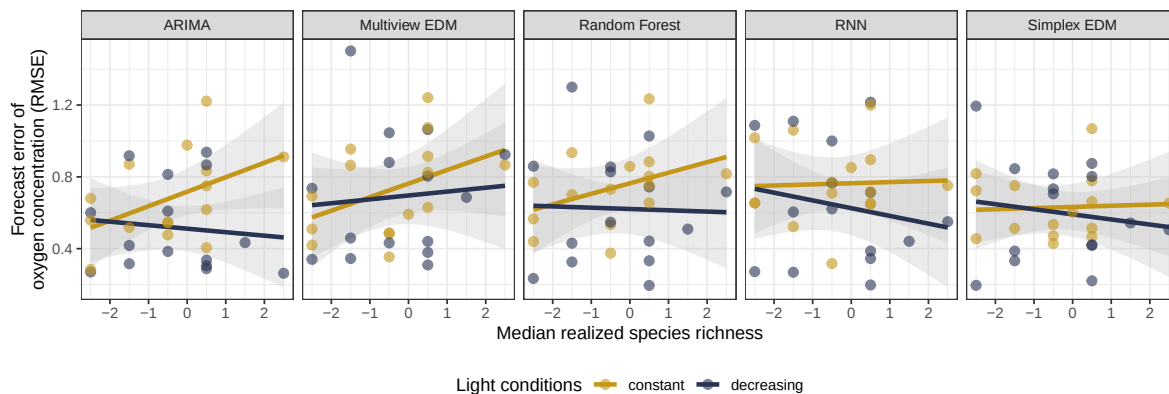


Figure 16: Relation between species richness and light and the forecast error of dissolved oxygen. Relationship between forecast error of oxygen and richness in constant light (yellow) and declining light (blue). Lines display the fit of the linear mixed model, shaded areas denote the 95% confidence intervals.

```

rownames(tab.bio)<- NULL

tab.bio$p.value <- ifelse(tab.bio$p.value<0.0001, "<0.0001",
                          sprintf("%.4f", round(tab.bio$p.value,4)))

tab.bio <- tab.bio %>%
  dplyr::mutate(Estimate = round(Estimate,3),
               SE = round(SE,3),
               Test.value = round(Test.value,3))

options(knitr.kable.NA = '-')
tab.bio %>%
  dplyr::filter(Test != "\\textit{F}") %>%
  mutate_all(linebreak, align="l") %>%
  kable("latex",
        booktabs = T, digits = 3,
        col.names = c("Response", "Method", "Covariate","Type", "Estimate", "DF", "SE", "Test","Value",
                      "Test.stat", "P.value"),
        caption = "Results of the regressions investigating the effects of species richness and light on oxygen concentration",
        align = c("llllrrrrrr"), escape = F,
        label = "BioTab") %>%

```

Table 7: Results of the regressions investigating the effects of species richness and light on the forecast errors of the community biomass (for different forecast methods).

Response	Method	Covariate	Type	Estimate	DF	SE	Test	Value	p-value
Forecast error of community biomass	Simplex EDM	Intercept	Fixed effect	0.668	21	0.111	<i>t</i>	6.036	<0.0001
		Richness		0.039	25	0.073	<i>t</i>	0.529	0.6017
		Light (decreasing)		-0.034	13	0.105	<i>t</i>	-0.329	0.7476
		Richness:Light		-0.046	13	0.075	<i>t</i>	-0.622	0.5445
		Composition	Random intercept	0.303	1	-	χ^2	4.702	0.0301
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	Multiview EDM	Intercept	Fixed effect	0.562	23	0.081	<i>t</i>	6.903	<0.0001
		Richness		0.003	25	0.055	<i>t</i>	0.056	0.9556
		Light (decreasing)		0.011	13	0.089	<i>t</i>	0.126	0.9020
		Richness:Light		-0.106	14	0.064	<i>t</i>	-1.661	0.1192
		Composition	Random intercept	0.186	1	-	χ^2	2.109	0.1465
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	ARIMA	Intercept	Fixed effect	0.520	6	0.072	<i>t</i>	7.223	0.0002
		Richness		0.003	22	0.046	<i>t</i>	0.068	0.9461
		Light (decreasing)		-0.009	6	0.101	<i>t</i>	-0.094	0.9279
		Richness:Light		-0.032	21	0.065	<i>t</i>	-0.489	0.6297
		Composition	Random intercept	0.000	1	-	χ^2	0.000	1.0000
		Incubator		0.055	1	-	χ^2	0.066	0.7966
	RNN	Intercept	Fixed effect	0.785	19	0.093	<i>t</i>	8.474	<0.0001
		Richness		0.049	25	0.059	<i>t</i>	0.819	0.4202
		Light (decreasing)		-0.015	13	0.079	<i>t</i>	-0.186	0.8554
		Richness:Light		-0.125	13	0.056	<i>t</i>	-2.228	0.0442
		Composition	Random intercept	0.274	1	-	χ^2	6.706	0.0096
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	Random Forest	Intercept	Fixed effect	0.573	9	0.089	<i>t</i>	6.474	0.0001
		Richness		-0.006	23	0.057	<i>t</i>	-0.108	0.9147
		Light (decreasing)		0.032	4	0.106	<i>t</i>	0.305	0.7748
		Richness:Light		-0.070	12	0.070	<i>t</i>	-1.001	0.3361
		Composition	Random intercept	0.169	1	-	χ^2	1.205	0.2722
		Incubator		0.057	1	-	χ^2	0.048	0.8270

```
collapse_rows(columns = c(1,2,4), latex_hline = 'custom', custom_latex_hline = c(1,4),
  valign = "middle")>%
kable_styling(font_size = 7)
```

```
rownames(tab.oxy) <- NULL

tab.oxy$p.value <- ifelse(tab.oxy$p.value < 0.0001, "<0.0001",
  sprintf("%.4f", round(tab.oxy$p.value, 4)))

tab.oxy <- tab.oxy %>%
  dplyr::mutate(
    Estimate = round(Estimate, 3),
    SE = round(SE, 3),
    Test.value = round(Test.value, 3)
  )

options(knitr.kable.NA = '-')
tab.oxy %>%
  dplyr::filter(Test != "\\textit{F}") %>%
```

```

mutate_all(linebreak, align = "l") %>%
kable(
  "latex",
  booktabs = T,
  digits = 3,
  col.names = c(
    "Response",
    "Method",
    "Covariate",
    "Type",
    "Estimate",
    "DF",
    "SE",
    "Test",
    "Value",
    "\\textit{p}-value"
  ),
  caption = "Results of the regressions investigating the effects of species
richness and light on the forecast errors of mean oxygen (for different
forecast methods).",
  align = c("llllrrrrcrr"),
  escape = F,
  label = "OxyTab"
) %>%
collapse_rows(
  columns = c(1, 2, 4),
  latex_hline = 'custom',
  custom_latex_hline = c(1, 4),
  valign = "middle"
) %>%
kable_styling(font_size = 7)

```

```

rownames(tab.aggr.bio) <- NULL

tab.aggr.bio$p.value <-
  ifelse(tab.aggr.bio$p.value < 0.0001,
    "<0.0001",
    sprintf("%.4f", round(tab.aggr.bio$p.value, 4)))

tab.aggr.bio <- tab.aggr.bio %>%
  dplyr::mutate(
    Estimate = round(Estimate, 3),
    SE = round(SE, 3),
    Test.value = round(Test.value, 3)
  )

options(knitr.kable.NA = '-')
tab.aggr.bio %>%
  dplyr::filter(Test != "\\textit{F}") %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",

```

Table 8: Results of the regressions investigating the effects of species richness and light on the forecast errors of mean oxygen (for different forecast methods).

Response	Method	Covariate	Type	Estimate	DF	SE	Test	Value	p-value
Forecast error of mean oxygen	Simplex EDM	Intercept	Fixed effect	0.632	26	0.066	t	9.601	<0.0001
		Richness		0.006	26	0.045	t	0.141	0.8891
		Light (decreasing)		-0.041	26	0.092	t	-0.451	0.6557
		Richness:Light		-0.035	26	0.064	t	-0.541	0.5928
		Composition	Random intercept	0.000	1	-	χ^2	0.000	1.0000
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	Multiview EDM	Intercept	Fixed effect	0.763	26	0.084	t	9.082	<0.0001
		Richness		0.075	26	0.058	t	1.303	0.2040
		Light (decreasing)		-0.066	13	0.116	t	-0.574	0.5756
		Richness:Light		-0.054	15	0.081	t	-0.665	0.5161
		Composition	Random intercept	0.048	1	-	χ^2	0.007	0.9315
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	ARIMA	Intercept	Fixed effect	0.717	26	0.067	t	10.763	<0.0001
		Richness		0.081	26	0.046	t	1.753	0.0914
		Light (decreasing)		-0.206	26	0.093	t	-2.219	0.0354
		Richness:Light		-0.100	26	0.065	t	-1.542	0.1351
		Composition	Random intercept	0.000	1	-	χ^2	0.000	1.0000
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	RNN	Intercept	Fixed effect	0.764	26	0.079	t	9.641	<0.0001
		Richness		0.006	26	0.055	t	0.115	0.9097
		Light (decreasing)		-0.139	26	0.110	t	-1.258	0.2197
		Richness:Light		-0.049	26	0.077	t	-0.639	0.5282
		Composition	Random intercept	0.000	1	-	χ^2	0.000	1.0000
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	Random Forest	Intercept	Fixed effect	0.764	26	0.074	t	10.293	<0.0001
		Richness		0.059	26	0.051	t	1.146	0.2621
		Light (decreasing)		-0.143	26	0.103	t	-1.386	0.1774
		Richness:Light		-0.066	26	0.072	t	-0.912	0.3704
		Composition	Random intercept	0.000	1	-	χ^2	0.000	1.0000
		Incubator		0.000	1	-	χ^2	0.000	1.0000

```

booktabs = T,
digits = 3,
col.names = c(
  "Response",
  "Method",
  "Covariate",
  "Type",
  "Estimate",
  "DF",
  "SE",
  "Test",
  "Value",
  "\\textit{p}-value"
),
caption = "Results of the regressions investigating whether the community
level measure total biomass is forecasted better than taxa based forecasts
(here: mean forecast error across all forecasted taxa). The models also
include species richness and light as explanatory variables, as well as all
pairwise interactions. The table reports the results for all considered
forecast methods.",
align = c("llllrrrrcrr"),
escape = F,
label = "AggregateTabBio"
) %>%
collapse_rows(
  columns = c(1, 2, 4),
  latex_hline = 'custom',
  custom_latex_hline = c(1, 4),
  valign = "middle"
) %>%
kable_styling(font_size = 7)

```

```

rownames(tab.aggr.oxy) <- NULL

tab.aggr.oxy$p.value <-
  ifelse(tab.aggr.oxy$p.value < 0.0001,
    "<0.0001",
    sprintf("%.4f", round(tab.aggr.oxy$p.value, 4)))

tab.aggr.oxy <- tab.aggr.oxy %>%
  dplyr::mutate(
    Estimate = round(Estimate, 3),
    SE = round(SE, 3),
    Test.value = round(Test.value, 3)
  )

options(knitr.kable.NA = '-')
tab.aggr.oxy %>%
  dplyr::filter(Test != "\\textit{F}") %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",

```

Table 9: Results of the regressions investigating whether the community level measure total biomass is forecasted better than taxa based forecasts (here: mean forecast error across all forecasted taxa). The models also include species richness and light as explanatory variables, as well as all pairwise interactions. The table reports the results for all considered forecast methods.

Response	Method	Covariate	Type	Estimate	DF	SE	Test	Value	<i>p</i> -value
Mean forecast error of abundances Versus Forecast error of community biomass	Simplex EDM	Intercept	Fixed effect	0.729	50	0.085	<i>t</i>	8.530	<0.0001
		Richness		0.009	42	0.053	<i>t</i>	0.165	0.8697
		Light (decreasing)		-0.074	50	0.119	<i>t</i>	-0.623	0.5364
		Measure (biomass)		-0.050	27	0.106	<i>t</i>	-0.471	0.6412
		Richness:Light		-0.071	26	0.065	<i>t</i>	-1.085	0.2878
		Richness:Measure		0.052	27	0.053	<i>t</i>	0.983	0.3341
		Light:Measure		0.028	27	0.146	<i>t</i>	0.194	0.8480
		Bottle	Random intercept	0.142	1	-	χ^2	1.090	0.2966
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	Multiview EDM	Intercept	Fixed effect	0.693	42	0.066	<i>t</i>	10.424	<0.0001
		Richness		0.023	35	0.043	<i>t</i>	0.529	0.5999
		Light (decreasing)		-0.070	42	0.092	<i>t</i>	-0.759	0.4520
		Measure (biomass)		-0.130	27	0.067	<i>t</i>	-1.956	0.0609
		Richness:Light		-0.100	26	0.056	<i>t</i>	-1.792	0.0848
		Richness:Measure		-0.019	27	0.033	<i>t</i>	-0.562	0.5785
		Light:Measure		0.083	27	0.092	<i>t</i>	0.901	0.3756
		Bottle	Random intercept	0.170	1	-	χ^2	6.384	0.0115
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	ARIMA	Intercept	Fixed effect	0.590	12	0.057	<i>t</i>	10.394	<0.0001
		Richness		0.009	32	0.034	<i>t</i>	0.255	0.8005
		Light (decreasing)		-0.135	12	0.079	<i>t</i>	-1.710	0.1139
		Measure (biomass)		-0.066	27	0.059	<i>t</i>	-1.124	0.2711
		Richness:Light		-0.049	21	0.043	<i>t</i>	-1.141	0.2667
		Richness:Measure		0.003	27	0.029	<i>t</i>	0.097	0.9233
		Light:Measure		0.120	27	0.081	<i>t</i>	1.483	0.1497
		Bottle	Random intercept	0.117	1	-	χ^2	3.308	0.0690
		Incubator		0.041	1	-	χ^2	0.106	0.7452
	RNN	Intercept	Fixed effect	0.892	49	0.073	<i>t</i>	12.301	<0.0001
		Richness		0.034	41	0.046	<i>t</i>	0.752	0.4563
		Light (decreasing)		-0.041	49	0.101	<i>t</i>	-0.411	0.6829
		Measure (biomass)		-0.097	27	0.088	<i>t</i>	-1.101	0.2806
		Richness:Light		-0.124	26	0.056	<i>t</i>	-2.211	0.0360
		Richness:Measure		0.037	27	0.044	<i>t</i>	0.849	0.4033
		Light:Measure		0.023	27	0.121	<i>t</i>	0.191	0.8498
		Bottle	Random intercept	0.133	1	-	χ^2	1.496	0.2213
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	Random Forest	Intercept	Fixed effect	0.739	44	0.071	<i>t</i>	10.366	<0.0001
		Richness		0.010	36	0.046	<i>t</i>	0.219	0.8279
		Light (decreasing)		-0.077	44	0.099	<i>t</i>	-0.773	0.4435
		Measure (biomass)		-0.157	27	0.076	<i>t</i>	-2.064	0.0487
		Richness:Light		-0.100	26	0.059	<i>t</i>	-1.695	0.1019
		Richness:Measure		0.004	27	0.038	<i>t</i>	0.103	0.9184
		Light:Measure		0.098	27	0.105	<i>t</i>	0.935	0.3581
		Bottle	Random intercept	0.169	1	-	χ^2	4.742	0.0294
		Incubator		0.000	1	-	χ^2	0.000	1.0000

```

booktabs = T,
digits = 3,
col.names = c(
  "Response",
  "Method",
  "Covariate",
  "Type",
  "Estimate",
  "DF",
  "SE",
  "Test",
  "Value",
  "\\textit{p}-value"
),
caption = "Results of the regressions investigating whether the community
level measure mean oxygen concentration is forecasted better than taxa based
forecasts (here: mean forecast error across all forecasted taxa). The models
also include species richness and light as explanatory variables, as well as
all pairwise interactions. The table reports the results for all considered
forecast methods.",
align = c("llllrrrrcrr"),
escape = F,
label = "AggregateTabOxy"
) %>%
collapse_rows(
  columns = c(1, 2, 4),
  latex_hline = 'custom',
  custom_latex_hline = c(1, 4),
  valign = "middle"
) %>%
kable_styling(font_size = 7)

```

```

rownames(em.tab) <- NULL

em.tab$p.value <- ifelse(em.tab$p.value < 0.0001, "<0.0001",
  sprintf("%.4f", round(em.tab$p.value, 4)))

em.tab <- em.tab %>%
  dplyr::mutate(
    estimate = round(estimate, 3),
    SE = round(SE, 3),
    t.ratio = round(t.ratio, 3)
  )

options(knitr.kable.NA = '-')
em.tab %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 3,
    col.names = c(

```

Table 10: Results of the regressions investigating whether the community level measure mean oxygen concentration is forecasted better than taxa based forecasts (here: mean forecast error across all forecasted taxa). The models also include species richness and light as explanatory variables, as well as all pairwise interactions. The table reports the results for all considered forecast methods.

Response	Method	Covariate	Type	Estimate	DF	SE	Test	Value	p-value
Mean forecast error of abundances Versus Forecast error of mean oxygen concentration	Simplex EDM	Intercept	Fixed effect	0.730	48	0.058	<i>t</i>	12.551	<0.0001
		Richness		0.011	40	0.037	<i>t</i>	0.312	0.7564
		Light (decreasing)		-0.076	48	0.081	<i>t</i>	-0.939	0.3523
		Measure (oxygen)		-0.088	27	0.069	<i>t</i>	-1.280	0.2116
		Richness:Light		-0.076	26	0.046	<i>t</i>	-1.665	0.1080
		Richness:Measure		0.016	27	0.034	<i>t</i>	0.455	0.6530
		Light:Measure		0.019	27	0.095	<i>t</i>	0.196	0.8462
		Bottle	Random intercept	0.113	1	-	χ^2	2.093	0.1480
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	Multiview EDM	Intercept	Fixed effect	0.688	50	0.068	<i>t</i>	10.177	<0.0001
		Richness		0.012	41	0.042	<i>t</i>	0.284	0.7775
		Light (decreasing)		-0.062	49	0.094	<i>t</i>	-0.660	0.5125
		Measure (oxygen)		0.081	27	0.083	<i>t</i>	0.968	0.3417
		Richness:Light		-0.079	26	0.052	<i>t</i>	-1.523	0.1399
		Richness:Measure		0.076	27	0.042	<i>t</i>	1.830	0.0784
		Light:Measure		-0.014	27	0.115	<i>t</i>	-0.123	0.9029
		Bottle	Random intercept	0.116	1	-	χ^2	1.239	0.2656
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	ARIMA	Intercept	Fixed effect	0.598	51	0.053	<i>t</i>	11.312	<0.0001
		Richness		0.027	43	0.033	<i>t</i>	0.824	0.4143
		Light (decreasing)		-0.147	51	0.073	<i>t</i>	-2.006	0.0502
		Measure (oxygen)		0.116	27	0.068	<i>t</i>	1.701	0.1004
		Richness:Light		-0.085	26	0.039	<i>t</i>	-2.154	0.0407
		Richness:Measure		0.046	27	0.034	<i>t</i>	1.349	0.1887
		Light:Measure		-0.053	27	0.094	<i>t</i>	-0.566	0.5762
		Bottle	Random intercept	0.074	1	-	χ^2	0.546	0.4601
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	RNN	Intercept	Fixed effect	0.885	51	0.064	<i>t</i>	13.766	<0.0001
		Richness		0.018	43	0.040	<i>t</i>	0.453	0.6531
		Light (decreasing)		-0.029	51	0.089	<i>t</i>	-0.325	0.7468
		Measure (oxygen)		-0.110	27	0.083	<i>t</i>	-1.331	0.1944
		Richness:Light		-0.092	26	0.048	<i>t</i>	-1.924	0.0654
		Richness:Measure		0.010	27	0.041	<i>t</i>	0.234	0.8171
		Light:Measure		-0.126	27	0.114	<i>t</i>	-1.106	0.2785
		Bottle	Random intercept	0.090	1	-	χ^2	0.541	0.4621
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	Random Forest	Intercept	Fixed effect	0.739	50	0.065	<i>t</i>	11.409	<0.0001
		Richness		0.009	42	0.040	<i>t</i>	0.235	0.8154
		Light (decreasing)		-0.076	50	0.090	<i>t</i>	-0.848	0.4007
		Measure (oxygen)		0.033	27	0.081	<i>t</i>	0.410	0.6853
		Richness:Light		-0.099	26	0.049	<i>t</i>	-1.994	0.0567
		Richness:Measure		0.066	27	0.040	<i>t</i>	1.629	0.1150
		Light:Measure		-0.080	27	0.111	<i>t</i>	-0.719	0.4784
		Bottle	Random intercept	0.108	1	-	χ^2	1.106	0.2930
		Incubator		0.000	1	-	χ^2	0.000	1.0000

Table 11: Table reporting the differences (i.e. the contrasts) between the marginal estimated mean of the forecast errors of the community level variables (total biomass and mean oxygen concentration) and of the mean taxa abundance forecast error. Also reported are the results of the tests of whether the estimated differences are different from zero.

Contrast	Method	Light	Estimate	SE	DF	<i>t</i> -Value	<i>p</i> -value
Species - Biomass	Simplex EDM	constant	0.070	0.103	27	0.678	0.5034
		decreasing	0.042	0.103	27	0.404	0.6893
	Multiview EDM	constant	0.123	0.065	27	1.899	0.0683
		decreasing	0.040	0.065	27	0.624	0.5381
	ARIMA	constant	0.067	0.057	27	1.174	0.2508
		decreasing	-0.053	0.057	27	-0.925	0.3630
	RNN	constant	0.111	0.085	27	1.299	0.2050
		decreasing	0.088	0.085	27	1.028	0.3131
	Random Forest	constant	0.158	0.074	27	2.141	0.0414
		decreasing	0.061	0.074	27	0.818	0.4204
Species - Oxygen	Simplex EDM	constant	0.094	0.067	27	1.404	0.1716
		decreasing	0.076	0.067	27	1.127	0.2696
	Multiview EDM	constant	-0.051	0.081	27	-0.634	0.5315
		decreasing	-0.037	0.081	27	-0.460	0.6495
	ARIMA	constant	-0.098	0.066	27	-1.482	0.1499
		decreasing	-0.045	0.066	27	-0.681	0.5014
	RNN	constant	0.114	0.081	27	1.414	0.1689
		decreasing	0.240	0.081	27	2.979	0.0061
	Random Forest	constant	-0.008	0.078	27	-0.100	0.9212
		decreasing	0.072	0.078	27	0.918	0.3670

```

"Contrast",
"Method",
"Light",
"Estimate",
"SE",
"DF",
"\\textit{t}-Value",
"\\textit{p}-value"
),
caption = "Table reporting the differences (i.e. the contrasts) between the
marginal estimated mean of the forecast errors of the community level variables
(total biomass and mean oxygen concentration) and of the mean taxa abundance
forecast error. Also reported are the results of the tests of whether the
estimated differences are different from zero.",
align = c("lllrrrcrr"),
escape = F,
label = "ContrastTab"
) %>%
collapse_rows(
  columns = c(1, 2),
  latex_hline = 'custom',
  custom_latex_hline = c(1, 2),
  valign = "middle"
) %>%
kable_styling(font_size = 7)

```

7.2 Relations of time series properties with forecast errors and with experimental conditions

```
simplex.forecasts.taxa <- complete.forecasts %>%
  dplyr::filter(Target != "Total_biomass", Target != "Mean_oxygen") %>%
  dplyr::select(
    "bottle",
    "richness",
    "composition",
    "incubator",
    "realized_richness",
    "light_treatment",
    "Target",
    "RMSE",
    "forecast.method",
    "mean_richness",
    "median_richness",
    "realized_richness_center",
    "median_richness_center",
    "mean_richness_center",
    "pe5",
    "cv.res",
    "acf_lag5"
  ) %>%
  dplyr::filter(!(Target %in% c("Total_biomass", "Mean_oxygen")), forecast.method ==
    "Simplex EDM") %>%
  mutate(cv.res.log10 = log10(cv.res))

## acf vs RMSE

m.acf.rmse <-
  lmer(
    RMSE ~ acf_lag5 * light_treatment + (1 |
      Target) + (1 | bottle) + (1 | incubator),
    simplex.forecasts.taxa
  )

## CV vs RMSE

m.cv.rmse <-
  lmer(
    RMSE ~ cv.res.log10 * light_treatment + (1 |
      Target) + (1 | bottle) + (1 | incubator),
    simplex.forecasts.taxa
  )

### pe vs rmse

m.pe.rmse <-
  lmer(
    RMSE ~ pe5 * light_treatment + (1 |
      Target) + (1 | bottle) + (1 | incubator),
    simplex.forecasts.taxa
  )
```

```

# tab

mods <- list(m.acf.rmse, m.cv.rmse, m.pe.rmse)
CoefNams <- c(
  "Autocorrelation\n(lag = 5)",
  "log10-transformed\nCoefficient of\nvariation)",
  "Permutation\nentropy"
)
CoefNamsShort <- c("ACF (lag 5)",
  "log10(CV)",
  "PE")
InteractionNams <- c("ACF:Light",
  "log10(CV):Light",
  "PE:Light")
CvAcfPeRmseTab <- data.frame()

for (i in 1:length(mods)) {
  m <- mods[[i]]
  r <- rand(m)

  t1 <- data.frame(
    MainCovariate = CoefNams[i],
    Covariate = c(
      "Intercept",
      CoefNamsShort[i],
      "Light (decreasing)",
      InteractionNams[i],
      "Target (taxa)",
      "Bottle",
      "Incubator"
    ),
    Type = c(
      rep("Fixed\neffect", 4),
      "Random\nintercept",
      "Random\nintercept",
      "Random\nintercept"
    ),
    Estimate = c(fixef(m), as.data.frame(VarCorr(m))$sdcor[1:3]),
    DF = c(round(summary(m)$coef[, 3, drop = FALSE], 0), r$Df[2:4]),
    SE = c(summary(m)$coef[, 2, drop = FALSE], NA, NA, NA),
    Test = c(rep("\\textit{t}", 4), "\\chi^2$", "\\chi^2$", "\\chi^2$"),
    Test.value = c(summary(m)$coef[, 4, drop = FALSE], r$LRT[2:4]),
    p.value = c(summary(m)$coef[, 5, drop = FALSE], r$`Pr(>Chisq)`[2:4])
  )

  t2 <- data.frame(
    MainCovariate = CoefNams[i],
    Covariate = c(CoefNamsShort[i], "Light (decreasing)", InteractionNams[i]),
    Type = "Sum of\nsquares",
    Estimate = anova(m)[, 1],
    DF = paste(anova(m)[, 3], round(anova(m)[, 4], 1), sep = ", "),
    SE = NA,
    Test = "\\textit{F}",

```

```

    Test.value = anova(m)[, 5],
    p.value = anova(m)[, 6]
  )

  CvAcfPeRmseTab <- rbind(CvAcfPeRmseTab, t1, t2)
}

rownames(CvAcfPeRmseTab) <- NULL

CvAcfPeRmseTab$p.value <-
  ifelse(CvAcfPeRmseTab$p.value < 0.0001,
    "<0.0001",
    sprintf("%.4f", round(CvAcfPeRmseTab$p.value, 4)))

CvAcfPeRmseTab <- CvAcfPeRmseTab %>%
  dplyr::mutate(
    Estimate = round(Estimate, 3),
    SE = round(SE, 3),
    Test.value = round(Test.value, 3)
  )

options(knitr.kable.NA = '-')
CvAcfPeRmseTab %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 3,
    col.names = c(
      "Main covariate",
      "Covariate",
      "Type",
      "Estimate",
      "DF",
      "SE",
      "Test",
      "Value",
      "\\textit{p}-value"
    ),
    caption = 'Results of linear mixed models investigating the relationship
between forecast error (response variable) and time series metrics. The
models were fitted separately for the explanatory variables "autocorrelation"
(at lag 5), "log10-transformed coefficient of variation", and "permutation
entropy". This table includes both the regression tables and the analysis of
variance (Type III) tables of the models. The models included the respective
time series metrics as a covariate, alongside the light conditions and their
interaction as fixed effects. The random intercepts were included for the
bottles, incubators and the targets (i.e. forecasted taxa). Forecast error
is based on forecasts computed with Simplex EDM.',
    align = c("lllrrrrcrr"),

```

Table 12: Results of linear mixed models investigating the relationship between forecast error (response variable) and time series metrics. The models were fitted separately for the explanatory variables "auto-correlation" (at lag 5), "log10-transformed coefficient of variation", and "permutation entropy". This table includes both the regression tables and the analysis of variance (Type III) tables of the models. The models included the respective time series metrics as a covariate, alongside the light conditions and their interaction as fixed effects. The random intercepts were included for the bottles, incubators and the targets (i.e. forecasted taxa). Forecast error is based on forecasts computed with Simplex EDM.

Main covariate	Covariate	Type	Estimate	DF	SE	Test	Value	p-value
Autocorrelation (lag = 5)	Intercept	Fixed effect	0.969	31	0.107	<i>t</i>	9.017	<0.0001
	ACF (lag 5)		-0.683	243	0.140	<i>t</i>	-4.894	<0.0001
	Light (decreasing)		0.016	51	0.123	<i>t</i>	0.135	0.8935
	ACF:Light		-0.016	242	0.201	<i>t</i>	-0.077	0.9383
	Target (taxa)	Random intercept	0.000	1	-	χ^2	64.108	<0.0001
	Bottle		0.269	1	-	χ^2	0.000	0.9995
	Incubator		0.062	1	-	χ^2	1.027	0.3107
	ACF (lag 5)	Sum of squares	6.293	1, 248.5	-	<i>F</i>	38.129	<0.0001
	Light (decreasing)		0.003	1, 50.8	-	<i>F</i>	0.018	0.8935
	ACF:Light		0.001	1, 241.7	-	<i>F</i>	0.006	0.9383
log10-transformed Coefficient of variation)	Intercept	Fixed effect	0.378	30	0.117	<i>t</i>	3.230	0.0030
	log10(CV)		-0.643	233	0.141	<i>t</i>	-4.560	<0.0001
	Light (decreasing)		0.120	17	0.122	<i>t</i>	0.981	0.3406
	log10(CV):Light		0.311	225	0.196	<i>t</i>	1.581	0.1153
	Target (taxa)	Random intercept	0.087	1	-	χ^2	49.352	<0.0001
	Bottle		0.253	1	-	χ^2	0.885	0.3468
	Incubator		0.101	1	-	χ^2	2.210	0.1371
	log10(CV)	Sum of squares	2.859	1, 225.1	-	<i>F</i>	17.033	<0.0001
	Light (decreasing)		0.162	1, 16.6	-	<i>F</i>	0.963	0.3406
	log10(CV):Light		0.419	1, 225.3	-	<i>F</i>	2.499	0.1153
Permutation entropy	Intercept	Fixed effect	-0.286	184	0.268	<i>t</i>	-1.068	0.2871
	PE		1.478	241	0.385	<i>t</i>	3.840	0.0002
	Light (decreasing)		0.442	194	0.352	<i>t</i>	1.255	0.2110
	PE:Light		-0.748	240	0.533	<i>t</i>	-1.404	0.1615
	Target (taxa)	Random intercept	0.121	1	-	χ^2	44.613	<0.0001
	Bottle		0.231	1	-	χ^2	2.442	0.1181
	Incubator		0.113	1	-	χ^2	1.964	0.1610
	PE	Sum of squares	2.177	1, 243	-	<i>F</i>	12.855	0.0004
	Light (decreasing)		0.267	1, 194.3	-	<i>F</i>	1.575	0.2110
	PE:Light		0.334	1, 239.8	-	<i>F</i>	1.972	0.1615

```

escape = F,
label = "CvAcfPeRmseTab"
) %>%
collapse_rows(
  columns = c(1, 3),
  latex_hline = 'custom',
  custom_latex_hline = c(1, 3),
  valign = "middle"
) %>%
kable_styling(font_size = 7)

```

```
## acf vs exp
```

```

m.acf.exp <-
  lmer(
    acf_lag5 ~ median_richness_center * light_treatment +

```

```

    (1 | Target) + (1 | bottle) + (1 | incubator),
    simplex.forecasts.taxa
  )

## cv vs exp

m.cv.exp <-
  lmer(
    cv.res.log10 ~ median_richness_center * light_treatment +
      (1 | Target) + (1 | bottle) + (1 | incubator),
    simplex.forecasts.taxa
  )

### pe vs rmse

m.pe.exp <-
  lmer(
    pe5 ~ median_richness_center * light_treatment +
      (1 | Target) + (1 | bottle) + (1 | incubator),
    simplex.forecasts.taxa
  )

# tab

mods <- list(m.acf.exp, m.cv.exp, m.pe.exp)
CoefNams <- c(
  "Autocorrelation\n(lag = 5)",
  "log10-transformed\nCoefficient of\nvariation)",
  "Permutation\nentropy"
)

EXPCvAcfPeTab <- data.frame()

for (i in 1:length(mods)) {
  m <- mods[[i]]
  r <- rand(m)

  t1 <- data.frame(
    Response = CoefNams[i],
    Covariate = c(
      "Intercept",
      "Richness",
      "Light (decreasing)",
      "Richness:Light",
      "Target (taxa)",
      "Bottle",
      "Incubator"
    ),
    Type = c(
      rep("Fixed\neffect", 4),
      "Random\nintercept",
      "Random\nintercept",
      "Random\nintercept"
    ),
  )
}

```

```

    Estimate = c(fixef(m), as.data.frame(VarCorr(m))$sdcor[1:3]),
    DF = c(round(summary(m)$coef[, 3, drop = FALSE], 0), r$Df[2:4]),
    SE = c(summary(m)$coef[, 2, drop = FALSE], NA, NA, NA),
    Test = c(rep("\\textit{t}", 4), "\\chi^2$", "\\chi^2$", "\\chi^2$"),
    Test.value = c(summary(m)$coef[, 4, drop = FALSE], r$LRT[2:4]),
    p.value = c(summary(m)$coef[, 5, drop = FALSE], r$`Pr(>Chisq)`[2:4])
  )

  t2 <- data.frame(
    Response = CoefNams[i],
    Covariate = c("Richness", "Light (decreasing)", "Richness:Light"),
    Type = "Sum of squares",
    Estimate = anova(m)[, 1],
    DF = paste(anova(m)[, 3], round(anova(m)[, 4], 1), sep = ", "),
    SE = NA,
    Test = "\\textit{F}",
    Test.value = anova(m)[, 5],
    p.value = anova(m)[, 6]
  )

  EXPCvAcfPeTab <- rbind(EXPCvAcfPeTab, t1, t2)
}

rownames(EXPCvAcfPeTab) <- NULL

EXPCvAcfPeTab$p.value <-
  ifelse(EXPCvAcfPeTab$p.value < 0.0001,
    "<0.0001",
    sprintf("%.4f", round(EXPCvAcfPeTab$p.value, 4)))

EXPCvAcfPeTab <- EXPCvAcfPeTab %>%
  dplyr::mutate(
    Estimate = round(Estimate, 3),
    SE = round(SE, 3),
    Test.value = round(Test.value, 3)
  )

options(knitr.kable.NA = '-')
EXPCvAcfPeTab %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 3,
    col.names = c(
      "Response",
      "Covariate",
      "Type",
      "Estimate",
      "DF",

```

```

      "SE",
      "Test",
      "Value",
      "\\textit{p}-value"
    ),
    caption = 'Results of linear mixed models investigating the the effect of
species richness and light on time series metrics (response variables). The
models were fitted separately for the response variables "autocorrelation"
(at lag 5), "log10-transformed coefficient of variation", and "permutation
entropy". This table includes both the regression tables and the analysis
of variance (Type III) tables of the models. The models included median
centered richness, light treatment and their interaction as fixed effects.
The random intercepts were included for the bottles, incubators and the
targets (i.e. forecasted taxa). Forecast error is based on forecasts
computed with Simplex EDM.',
    align = c("lllrrrrcrr"),
    escape = F,
    label = "EXPCvAcfPeTab"
  ) %>%
collapse_rows(
  columns = c(1, 3),
  latex_hline = 'custom',
  custom_latex_hline = c(1, 3),
  valign = "middle"
) %>%
kable_styling(font_size = 7)

```

7.3 Structural equation model

```

semDD <- complete.forecasts %>%
  dplyr::filter(Target != "Total_biomass", Target != "Mean_oxygen") %>%
  ungroup() %>%
  mutate(light_treatment_num = ifelse(light_treatment == "constant", 0, 1))

semDD <- data.frame(semDD)

## create random effect variable
semDD$group <- paste0(semDD$Target, "_",
                      semDD$bottle, "_",
                      semDD$incubator)

semDD.simplex <- semDD %>%
  dplyr::filter(forecast.method == "Simplex EDM")

semDD.rf <- semDD %>%
  dplyr::filter(forecast.method == "Random Forest")

semDD.mv <- semDD %>%
  dplyr::filter(forecast.method == "Multiview EDM")

```

Table 13: Results of linear mixed models investigating the the effect of species richness and light on time series metrics (response variables). The models were fitted separately for the response variables "autocorrelation" (at lag 5), "log10-transformed coefficient of variation", and "permutation entropy". This table includes both the regression tables and the analysis of variance (Type III) tables of the models. The models included median centered richness, light treatment and their interaction as fixed effects. The random intercepts were included for the bottles, incubators and the targets (i.e. forecasted taxa). Forecast error is based on forecasts computed with Simplex EDM.

Response	Covariate	Type	Estimate	DF	SE	Test	Value	p-value
Autocorrelation (lag = 5)	Intercept	Fixed effect	0.426	16	0.054	t	7.886	<0.0001
	Richness		-0.011	22	0.023	t	-0.474	0.6405
	Light (decreasing)		0.135	6	0.047	t	2.886	0.0296
	Richness:Light		0.030	20	0.032	t	0.944	0.3565
	Target (taxa)	Random intercept	0.088	1	-	χ^2	54.521	<0.0001
	Bottle		0.152	1	-	χ^2	6.773	0.0093
	Incubator		0.025	1	-	χ^2	0.047	0.8286
	Richness	Sum of squares	0.003	1, 20.6	-	F	0.068	0.7963
	Light (decreasing)		0.398	1, 5.7	-	F	8.329	0.0296
	Richness:Light		0.043	1, 19.8	-	F	0.891	0.3565
log10-transformed Coefficient of variation)	Intercept	Fixed effect	-0.428	12	0.063	t	-6.761	<0.0001
	Richness		0.004	235	0.015	t	0.282	0.7781
	Light (decreasing)		0.129	235	0.028	t	4.671	<0.0001
	Richness:Light		-0.022	236	0.021	t	-1.077	0.2825
	Target (taxa)	Random intercept	0.000	1	-	χ^2	90.545	<0.0001
	Bottle		0.219	1	-	χ^2	0.000	1.0000
	Incubator		0.000	1	-	χ^2	0.000	1.0000
	Richness	Sum of squares	0.020	1, 237.9	-	F	0.426	0.5148
	Light (decreasing)		1.031	1, 234.9	-	F	21.822	<0.0001
	Richness:Light		0.055	1, 236.3	-	F	1.160	0.2825
Permutation entropy	Intercept	Fixed effect	0.642	14	0.023	t	27.968	<0.0001
	Richness		0.013	235	0.006	t	2.247	0.0256
	Light (decreasing)		-0.032	6	0.019	t	-1.708	0.1375
	Richness:Light		0.004	235	0.008	t	0.544	0.5869
	Target (taxa)	Random intercept	0.000	1	-	χ^2	72.825	<0.0001
	Bottle		0.067	1	-	χ^2	0.000	1.0000
	Incubator		0.022	1	-	χ^2	3.635	0.0566
	Richness	Sum of squares	0.099	1, 237.3	-	F	14.032	0.0002
	Light (decreasing)		0.020	1, 6.1	-	F	2.917	0.1375
	Richness:Light		0.002	1, 234.9	-	F	0.296	0.5869

```

semDD.rnn <- semDD %>%
  dplyr::filter(forecast.method == "RNN")

semDD.ar <- semDD %>%
  dplyr::filter(forecast.method == "ARIMA")

model <-
  "
  RMSE ~ median_richness + cv.res + pe5 + acf_lag5
  cv.res ~ median_richness
  pe5 ~ median_richness
  acf_lag5 ~ median_richness
  cv.res ~~ pe5
  "

# semDD.simplex, semDD.rf, semDD.mv, semDD.rnn, semDD.ar
out <- sem(model, data = semDD.simplex, group = "light_treatment")

### Survey Design Object
des.card <- svydesign(ids = ~ group,
  nest = T,
  data = semDD.simplex)

### Post-Processing
fit.out.survey <- lavaan.survey(out, des.card, estimator = "MLM")

SEMsum <- summary(fit.out.survey, standardized = T, rsquare = T)

SEMtab <- SEMsum$pe %>%
  dplyr::filter(rhs != "") %>%
  dplyr::mutate(
    Light = ifelse(group == 1, "Constant light", "Decreasing light"),
    lhs = case_when(
      lhs == "RMSE" ~ "Forecast error",
      lhs == "cv.res" ~ "Coefficient of variation",
      lhs == "pe5" ~ "Permutation entropy",
      lhs == "acf_lag5" ~ "Autocorrelation (lag 5)",
      lhs == "median_richness" ~ "Species richness"
    ),
    rhs = case_when(
      rhs == "RMSE" ~ "Forecast error",
      rhs == "cv.res" ~ "Coefficient of variation",
      rhs == "pe5" ~ "Permutation entropy",
      rhs == "acf_lag5" ~ "Autocorrelation (lag 5)",
      rhs == "median_richness" ~ "Species richness"
    )
  )

SEMtabR2 <- SEMtab %>%
  dplyr::filter(op == "r2") %>%
  dplyr::select(Light, Variable = lhs, est)

SEMtab <- SEMtab %>%

```

```

dplyr::filter(op != "r2",
              lhs != rhs) %>%
dplyr::mutate(op = ifelse(op == "~", "$\\sim$", "$\\sim$$\\sim$")) %>%
dplyr::mutate(term = paste0(lhs, op, rhs)) %>%
dplyr::select(Light, term, est, se, z, pvalue, std.all)

```

```

rownames(SEMtab) <- NULL

SEMtab$pvalue <- ifelse(SEMtab$pvalue < 0.0001, "<0.0001",
                      sprintf("%.4f", round(SEMtab$pvalue, 4)))

SEMtab <- SEMtab %>%
  dplyr::mutate(
    est = round(est, 3),
    se = round(se, 3),
    z = round(z, 3),
    std.all = round(std.all, 3)
  )

options(knitr.kable.NA = '-')

SEMtab %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 3,
    col.names = c(
      "Light conditions",
      "Term",
      "Estimate",
      "SE",
      "\\textit{z}-value",
      "\\textit{p}-value",
      "Stand. est."
    ),
    caption = 'Table of path coefficients of the structural equation model.
The last column reports the standardized coefficients.',
    align = c("llrrrrr"),
    escape = F,
    label = "SEMtab"
  ) %>%
  collapse_rows(
    columns = 1,
    latex_hline = 'custom',
    custom_latex_hline = 1,
    valign = "middle"
  ) %>%
  kable_styling(font_size = 7)

```

```

rownames(SEMtabR2) <- NULL

```

Table 14: Table of path coefficients of the structural equation model. The last column reports the standardized coefficients.

Light conditions	Term	Estimate	SE	z-value	p-value	Stand. est.
Constant light	Forecast error~Species richness	0.015	0.031	0.469	0.6391	0.035
	Forecast error~Coefficient of variation	-0.602	0.177	-3.412	0.0006	-0.416
	Forecast error~Permutation entropy	0.085	0.659	0.128	0.8978	0.017
	Forecast error~Autocorrelation (lag 5)	-0.695	0.112	-6.230	<0.0001	-0.368
	Coefficient of variation~Species richness	0.005	0.022	0.243	0.8084	0.018
	Permutation entropy~Species richness	0.010	0.007	1.459	0.1446	0.117
	Autocorrelation (lag 5)~Species richness	-0.018	0.019	-0.948	0.3429	-0.081
	Coefficient of variation~Permutation entropy	-0.032	0.008	-4.110	<0.0001	-0.784
Decreasing light	Forecast error~Species richness	-0.077	0.035	-2.224	0.0261	-0.220
	Forecast error~Coefficient of variation	-0.095	0.173	-0.548	0.5834	-0.072
	Forecast error~Permutation entropy	0.430	0.569	0.756	0.4495	0.092
	Forecast error~Autocorrelation (lag 5)	-0.350	0.143	-2.455	0.0141	-0.185
	Coefficient of variation~Species richness	-0.030	0.026	-1.150	0.2500	-0.111
	Permutation entropy~Species richness	0.014	0.007	2.186	0.0288	0.191
	Autocorrelation (lag 5)~Species richness	0.010	0.017	0.592	0.5536	0.053
	Coefficient of variation~Permutation entropy	-0.022	0.006	-3.968	<0.0001	-0.628

```
SEMrtabR2 <- SEMtabR2 %>%
  dplyr::mutate(est = round(est, 4))

options(knitr.kable.NA = '-')

SEMrtabR2 %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 4,
    col.names = c("Light conditions", "Variable", "R-squared"),
    caption = 'Achieved coefficients of determination (R-squared)
for the structural equation model.',
    align = c("llr"),
    escape = F,
    label = "SEMrtabR2"
  ) %>%
  collapse_rows(
    columns = 1,
    latex_hline = 'custom',
    custom_latex_hline = 1,
    valign = "middle"
  ) %>%
  kable_styling(font_size = 8)
```

8 Robustness analyses

8.1 Effects of chosen richness variable and forecast method

Table 15: Achieved coefficients of determination (R-squared) for the structural equation model.

Light conditions	Variable	R-squared
Constant light	Forecast error	0.3222
	Coefficient of variation	0.0003
	Permutation entropy	0.0137
	Autocorrelation (lag 5)	0.0066
Decreasing light	Forecast error	0.0972
	Coefficient of variation	0.0124
	Permutation entropy	0.0365
	Autocorrelation (lag 5)	0.0028

```

formula <-
  as.formula(
    paste0(
      "RMSE ~ ",
      "richness_var",
      "*light_treatment + (1|Target) + (1|bottle) + (1|incubator)"
    )
  )
rich_vars2 <-
  list(
    "realized_richness_center",
    "median_richness_center",
    "mean_richness_center",
    "richness"
  )
fore.methods <-
  list("Multiview EDM", "Simplex EDM", "RNN", "ARIMA", "Random Forest")

tab <-
  robustnessFunction(rich_vars2, fore.methods, complete.forecasts,
    formula, phases = F)

residuals <- do.call("rbind", map(tab, 2))
tab.robust.main <- do.call("rbind", map(tab, 1))

tab.robust.main <- tab.robust.main %>%
  mutate(
    method = ifelse(method == "RNN", "Recurrent Neural Networks", method),
    richness_var = case_when(
      richness_var == "mean_richness_center" ~ "Mean richness",
      richness_var == "median_richness_center" ~
        "Median richness",
      richness_var == "realized_richness_center" ~
        "Minimal richness",
      richness_var == "richness" ~ "Planned richness"
    ),
    term = case_when(
      term == "(Intercept)" ~ "Intercept",
      term == "light_treatmentdecreasing" ~ "Light conditions (decreasing)",
      term == "richness_var" ~ "Species richness",
  
```

```

    term == "richness_var:light_treatmentdecreasing" ~
      "Interaction: light conditions and species richness"
  ),
  term = factor(
    term,
    levels = c(
      "Intercept",
      "Light conditions (decreasing)",
      "Species richness",
      "Interaction: light conditions and species richness"
    )
  )
)

tab.robust.main %>%
  ggplot(aes(richness_var, estimate, col = method)) +
  geom_hline(yintercept = 0, linetype = 2) +
  geom_errorbar(
    aes(ymin = conf.low, ymax = conf.high),
    width = 0.2,
    position = position_dodge(width = 0.4)
  ) +
  geom_point(position = position_dodge(width = 0.4)) +
  facet_wrap(~ term, scales = "free") +
  labs(col = "Forecast method", x = "Used species richness variable", y =
    "Coefficient estimates + 95% CI") +
  scale_color_brewer(palette = "Dark2", aesthetics = c("fill", "colour")) +
  theme_bw() +
  theme(
    legend.position = "bottom",
    strip.background = element_rect(fill = "white"),
    strip.text = element_text(color = "black", face = "bold")
  )

```

8.2 Effect of used detrending method

```

dd_list$densities.int.detrend.complete %>%
  dplyr::filter(!(
    species %in% c(
      "Algae",
      "Debris",
      "Didinium nasutum",
      "Digested algae",
      "Small unidentified",
      "Colpidium vacuoles"
    )
  ),
  ForecastTarget == T) %>%
  mutate(bottle = gsub("b_", "Bottle ", bottle)) %>%
  ggplot(aes(day, density.seg, col = species)) +
  geom_line(linewidth = 0.6) +

```

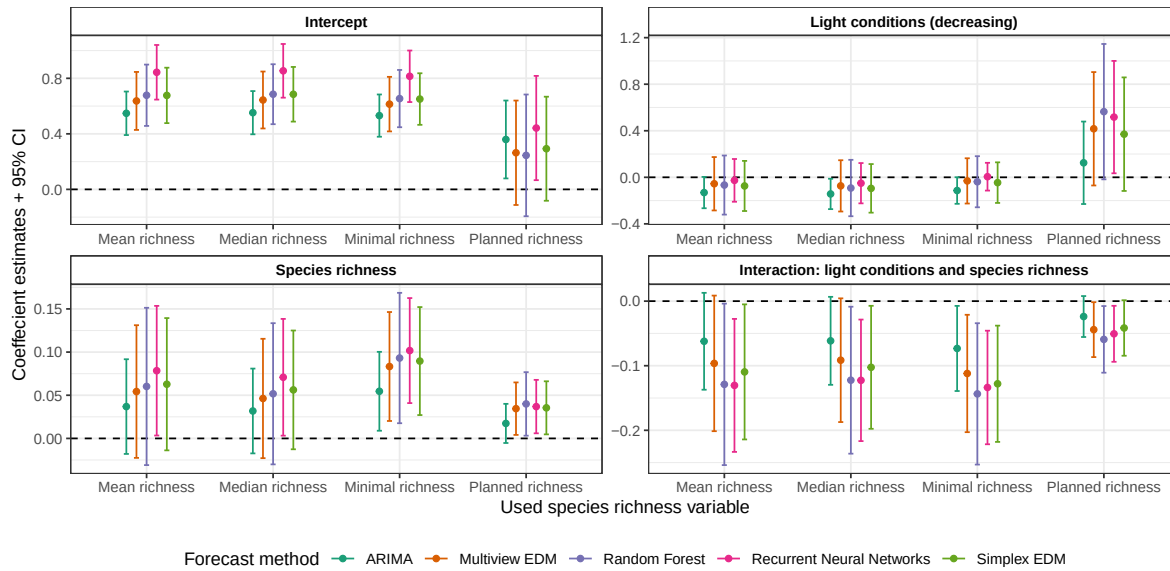


Figure 17: Robustness analysis figure regarding the chosen species richness variable and the forecast method. The figure depicts the coefficient estimates for regression between the target abundance forecast errors (response) and the covariates "species richness" and "light conditions", as specified by the panel labels. The dots are the estimates and the corresponding error-bars are the 95% CI. The colour specifies the used forecast method and the x-axis specifies the used species richness variable.

```
facet_wrap2( ~ bottle, ncol = 4, strip = strip) +
theme_bw() +
theme(
  legend.position = c(0.75, 0.055),
  panel.grid = element_blank(),
  legend.text = element_text(size = 12),
  axis.text = element_text(size = 11),
  axis.title = element_text(size = 16),
  strip.text = element_text(size = 12),
  # legend.key.size = unit(3, "line"),
  # strip.text = element_text(colour = "white"),
) +
scale_color_manual(values = unname(kelly()[2:16])) +
labs(col = "", y = "Standardized density",
  x = "Sampling day") +
guides(col = guide_legend(ncol = 3, byrow = TRUE))
```

```
detrend.compare.complete <-
full_join(
  seg.complete.forecasts %>%
    dplyr::mutate(detrend = "segmented"),
  complete.forecasts %>%
    dplyr::mutate(detrend = "simple")
)

detrend.compare.complete2 <-
full_join(
```



Figure 18: *Time series detrending based on the segmented temporal regression.* The *transformed* biotic density time series that were used as forecast targets, shown in the different bottles (i.e. the panels). The time series are coloured by forecast target. The background colour of the subpanel strips indicates whether the bottles were in a constant (orange) or in a decreasing (blue) light environment.

```

complete.forecasts %>%
  dplyr::select(
    bottle,
    Target,
    richness,
    composition,
    realized_richness,
    light_treatment,
    RMSE,
    forecast.method
  ),
seg.complete.forecasts %>%
  dplyr::select(
    bottle,
    Target,
    richness,
    composition,
    realized_richness,
    light_treatment,
    RMSE,
    forecast.method
  ) %>%
  dplyr::rename(RMSE_seg = RMSE)
)

```

```

tab <-
  robustnessFunction(rich_vars2,
    fore.methods,
    seg.complete.forecasts,
    formula,
    phases = F)

tab.detrending <- do.call("rbind", map(tab, 1)) %>%
  mutate(
    method = ifelse(method == "RNN", "Recurrent Neural Networks", method),
    richness_var = case_when(
      richness_var == "mean_richness_center" ~ "Mean richness",
      richness_var == "median_richness_center" ~
        "Median richness",
      richness_var == "realized_richness_center" ~
        "Minimal richness",
      richness_var == "richness" ~ "Planned richness"
    ),
    term = case_when(
      term == "(Intercept)" ~ "Intercept",
      term == "light_treatmentdecreasing" ~ "Light conditions (decreasing)",
      term == "richness_var" ~ "Species richness",
      term == "richness_var:light_treatmentdecreasing" ~
        "Interaction: light conditions and species richness"
    ),
    term = factor(
      term,
      levels = c(

```

```

    "Intercept",
    "Light conditions (decreasing)",
    "Species richness",
    "Interaction: light conditions and species richness"
  )
)
)

tab.detrending %>%
  ggplot(aes(richness_var, estimate, col = method)) +
  geom_hline(yintercept = 0, linetype = 2) +
  geom_errorbar(
    aes(ymin = conf.low, ymax = conf.high),
    width = 0.2,
    position = position_dodge(width = 0.4)
  ) +
  geom_point(position = position_dodge(width = 0.4)) +
  facet_wrap(~ term, scales = "free") +
  labs(col = "Forecast method", x = "Used species richness variable", y =
    "Coefficient estimates + 95% CI") +
  scale_color_brewer(palette = "Dark2", aesthetics = c("fill", "colour")) +
  theme_bw() +
  theme(
    legend.position = "bottom",
    strip.background = element_rect(fill = "white"),
    strip.text = element_text(color = "black", face = "bold")
  )
)

```

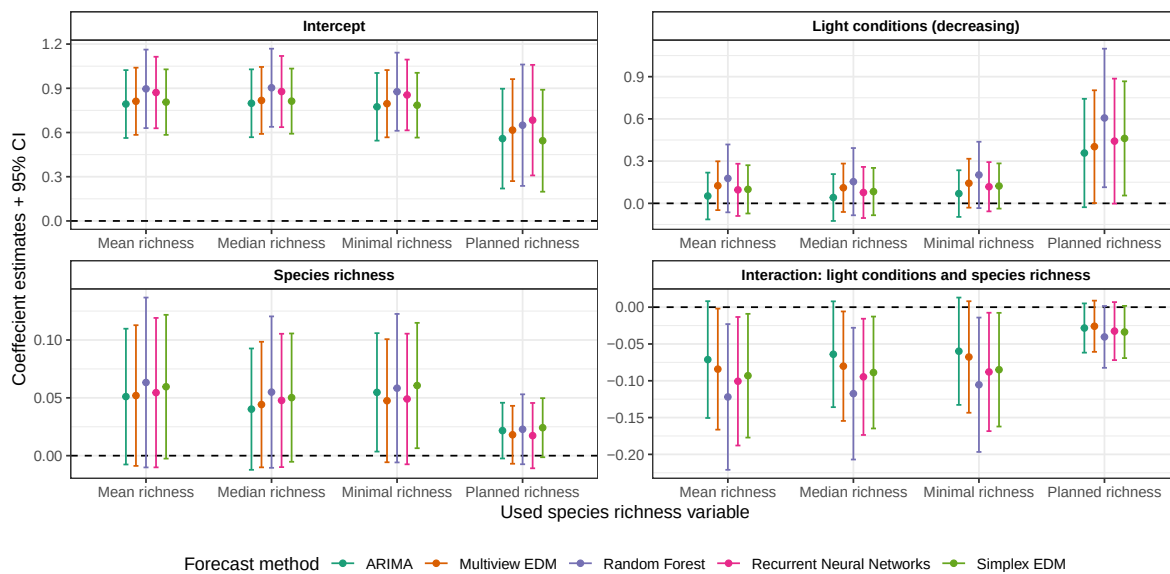


Figure 19: Robustness analysis figure for when the time series were detrended using the **segmented regression** approach. The figure depicts the coefficient estimates for regression between the target abundance forecast errors (response) and the covariates "species richness" and "light conditions", as specified by the panel labels. The dots are the estimates and the corresponding error-bars are the 95% CI. The colour specifies the used forecast method and the x-axis specifies the used species richness variable.

8.3 Species Effects on results

```
targets_excl <- unique(complete.forecasts$Target)
targets_excl <-
  targets_excl[!(targets_excl %in% c("Total_biomass", "Mean_oxygen"))]

tab <-
  robustnessFunctionSpecies(rich_vars2,
                            "Simplex EDM",
                            complete.forecasts,
                            formula,
                            targets_excl,
                            phases = F)

tab.speciesEffects <- do.call("rbind", tab) %>%
  mutate(
    richness_var = case_when(
      richness_var == "mean_richness_center" ~ "Mean richness",
      richness_var == "median_richness_center" ~
        "Median richness",
      richness_var == "realized_richness_center" ~
        "Minimal richness",
      richness_var == "richness" ~ "Planned richness"
    ),
    term = case_when(
      term == "(Intercept)" ~ "Intercept",
      term == "light_treatmentdecreasing" ~ "Light conditions (decreasing)",
      term == "richness_var" ~ "Species richness",
      term == "richness_var:light_treatmentdecreasing" ~
        "Interaction: light conditions and species richness"
    ),
    term = factor(
      term,
      levels = c(
        "Intercept",
        "Light conditions (decreasing)",
        "Species richness",
        "Interaction: light conditions and species richness"
      )
    )
  )

tab.speciesEffects %>%
  ggplot(aes(richness_var, estimate, col = ExcludedTarget)) +
  geom_hline(yintercept = 0, linetype = 2) +
  geom_errorbar(
    aes(ymin = conf.low, ymax = conf.high),
    width = 0.2,
    position = position_dodge(width = 0.5)
  ) +
  geom_point(position = position_dodge(width = 0.5)) +
  facet_wrap(~ term, scales = "free") +
```

```

labs(col = "Excluded\\ntaxa", x = "Used species richness variable",
     y = "Coefficient estimates + 95% CI") +
# scale_color_brewer(palette = "Dark2", aesthetics = c("fill", "colour")) +
scale_color_manual(values = unname(kelly()[2:15])) +
theme_bw() +
theme(
  legend.position = "bottom",
  strip.background = element_rect(fill = "white"),
  strip.text = element_text(color = "black", face = "bold")
)

```

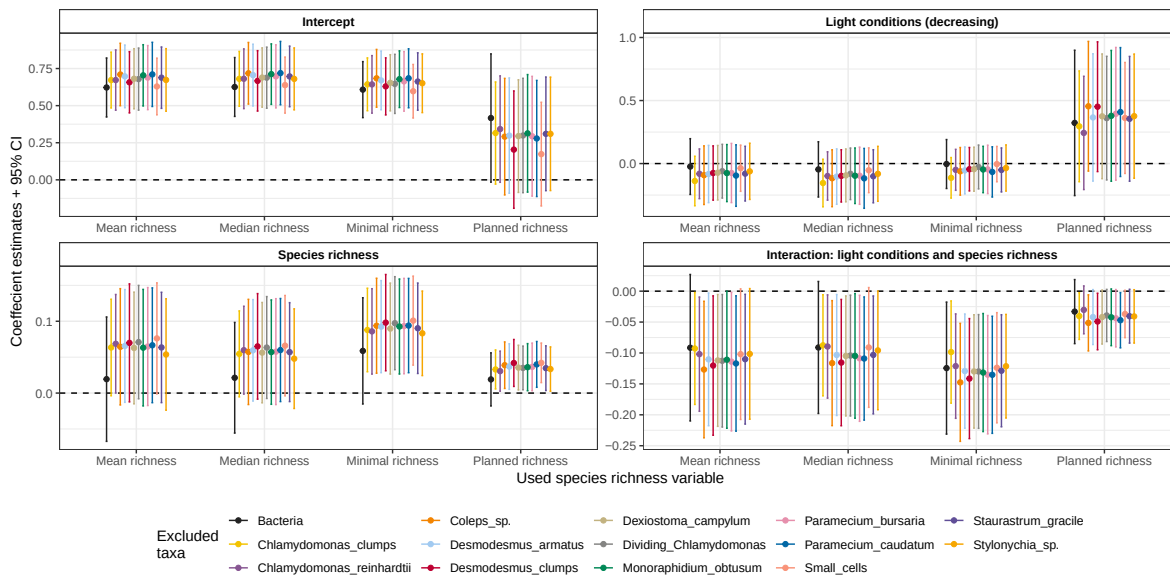


Figure 20: Robustness analysis figure regarding the sensitivity of the results on specific forecasted taxa. The figure depicts the coefficient estimates for regression between the target abundance forecast errors (response) and the covariates "species richness" and "light conditions" (as specified by the panel labels) for when the specified taxa were excluded (i.e. not forecasted). The dots are the estimates and the corresponding error-bars are the 95% CI. The colour specifies the taxa that was excluded and the x-axis specifies the used species richness variable. The forecasts are based on the Simplex EDM approach.

9 Further figures mentioned in the discussion

9.1 Insurance effect of diversity and its consequences for forecastability

```

stripColdd <- dd_list$densities %>%
  ungroup() %>%
  dplyr::select(bottle, light_treatment) %>%
  distinct() %>%
  mutate(stripCol = ifelse(light_treatment == "constant", "#f5d682", "#bdc5de"))

# Only colour strips in x-direction
strip <-

```

```

strip_themed(background_x = elem_list_rect(fill = stripColdd$stripCol))

dd_list$densities %>%
  dplyr::filter((
    species %in% c(
      "Coleps sp.",
      "Paramecium bursaria",
      "Colpidium striatum",
      "Loxocephalus sp.",
      "Paramecium caudatum",
      "Stylonychia sp.",
      "Dexiostoma campylum",
      "Euplotes daidaleos"
    )
  )) %>%
  mutate(bottle = gsub("b_", "Bottle ", bottle)) %>%
  ggplot(aes(day, log10(density), col = species)) +
  geom_line(linewidth = 0.6) +
  facet_wrap2( ~ bottle, ncol = 4, strip = strip) +
  theme_bw() +
  theme(
    legend.position = c(0.75, 0.055),
    panel.grid = element_blank(),
    legend.text = element_text(size = 12),
    axis.text = element_text(size = 11),
    axis.title = element_text(size = 16),
    strip.text = element_text(size = 12),

  ) +
  scale_color_manual(values = unname(kelly()[2:16])) +
  labs(col = "", y = expression(paste("log"[10], "(Density) (cells ", mL ^
    {
      -1
    }, ")")),
    x = "Sampling day") +
  guides(col = guide_legend(ncol = 3, byrow = TRUE))

```

```

stripColdd <- dd_list$densities %>%
  ungroup() %>%
  dplyr::select(bottle, light_treatment) %>%
  distinct() %>%
  mutate(stripCol = ifelse(light_treatment == "constant", "#f5d682", "#bdc5de"))

# Only colour strips in x-direction
strip <-
  strip_themed(background_x = elem_list_rect(fill = stripColdd$stripCol))

dd_list$densities %>%
  dplyr::filter((
    species %in% c(
      "Chlamydomonas clumps",
      "Chlamydomonas reinhardtii",
      "Cosmarium botrytis",

```

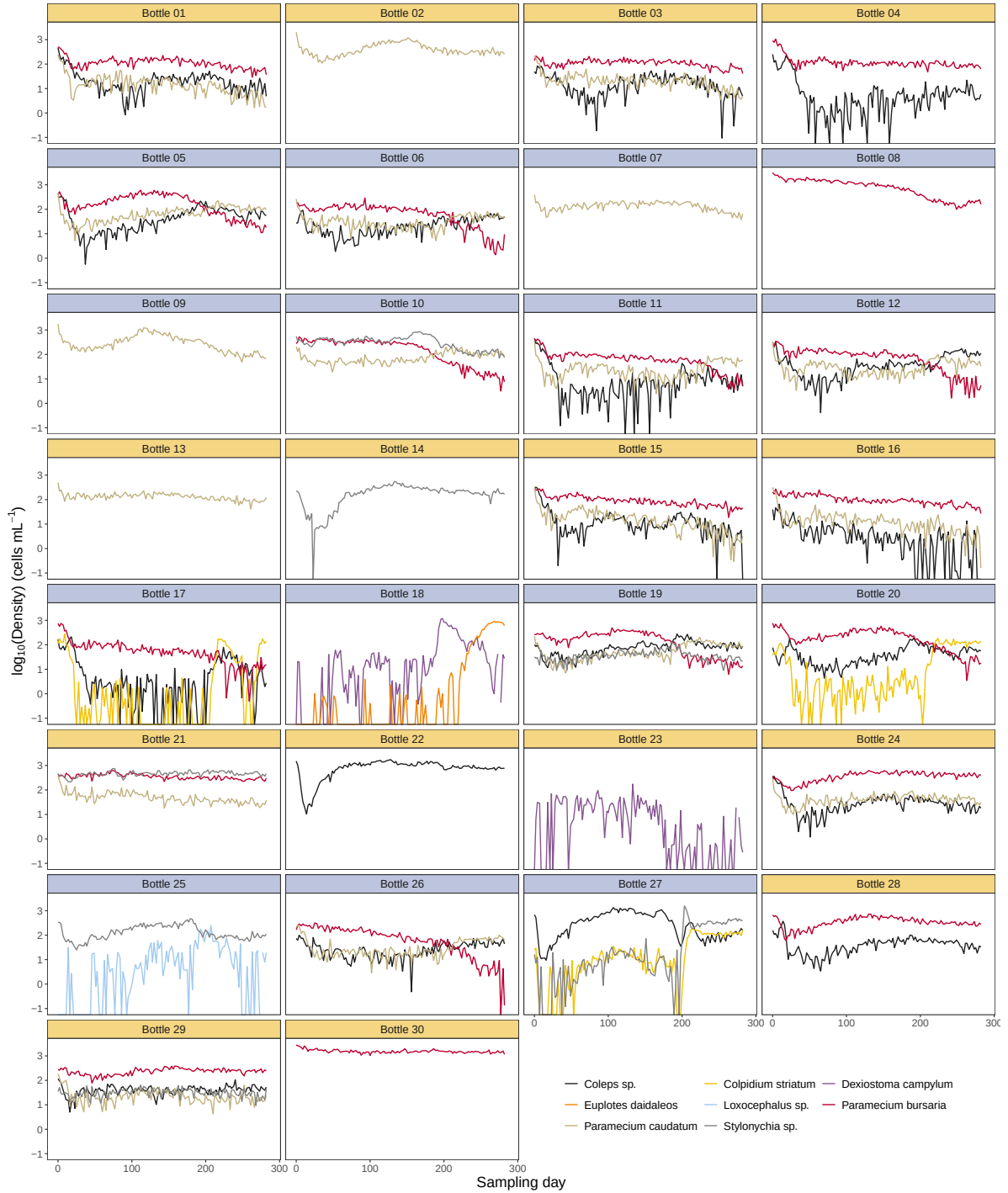


Figure 21: Time series of ciliate densities in the 30 experimental communities. Time series are coloured by ciliate species. The background colour of the subpanel strips indicates whether bottles were maintained in constant light (orange) or in decreasing light (blue).

```

      "Desmodesmus armatus",
      "Desmodesmus clumps",
      "Dividing Chlamydomonas",
      "Small cells",
      "Staurostrum gracile",
      "Monoraphidium obtusum",
      "Staurostrum polytrichum"
    )
  }) %>%
  mutate(bottle = gsub("b_", "Bottle ", bottle)) %>%
  ggplot(aes(day, log10(density), col = species)) +
  geom_line(linewidth = 0.6) +
  facet_wrap2( ~ bottle, ncol = 4, strip = strip) +
  theme_bw() +
  theme(
    legend.position = c(0.75, 0.055),
    panel.grid = element_blank(),
    legend.text = element_text(size = 12),
    axis.text = element_text(size = 11),
    axis.title = element_text(size = 16),
    strip.text = element_text(size = 12),
  ) +
  scale_color_manual(values = unname(kelly()[2:16])) +
  labs(col = "", y = expression(paste("log"[10], "(Density) (cells ", mL ^
    {
      -1
    }, ")")),
    x = "Sampling day") +
  guides(col = guide_legend(ncol = 3, byrow = TRUE))

```

9.2 Coefficient of variation of aggregate ecosystem properties

BIOMASS CV VERSUS EXPERIMENTAL CONDITIONS

```

complete.forecasts.pe.cv2.biomass <- complete.forecasts %>%
  ungroup() %>%
  dplyr::filter(Target == "Total_biomass", forecast.method == "Simplex EDM") %>%
  dplyr::select(
    bottle,
    median_richness,
    richness,
    light_treatment,
    composition,
    cv.res,
    median_richness_center,
    incubator
  ) %>%
  dplyr::mutate(log10.cv.res = log10(cv.res))

```

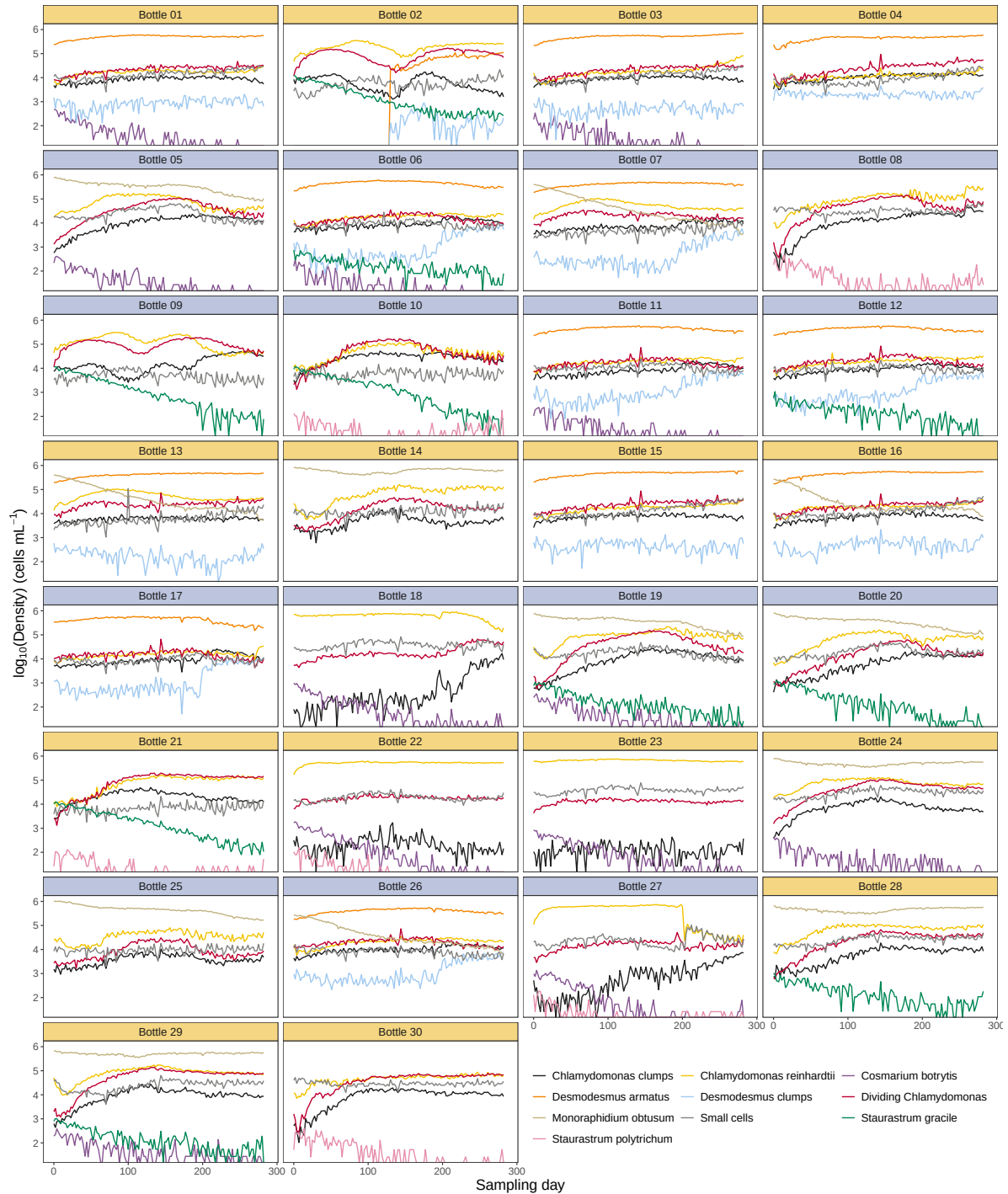


Figure 22: Time series of algae densities in the 30 experimental communities. Time series are coloured by algal species. The background colour of the subpanel strips indicates whether bottles were maintained in constant light (orange) or in decreasing light (blue).

```

m.cv.bio <-
  lmer(
    log10.cv.res ~ median_richness_center * light_treatment +
      (1 | composition) + (1 | incubator),
    complete.forecasts.pe.cv2.biomass
  )

newdat.m.cv.bio <- with(
  complete.forecasts.pe.cv2.biomass,
  expand.grid(
    median_richness_center = seq(
      min(median_richness_center),
      max(median_richness_center),
      length.out = 100
    ),
    light_treatment = unique(light_treatment)
  )
)

newdat.m.cv.bio <-
  conf_interval(m.cv.bio,
    newdat.m.cv.bio,
    "log10.cv.res",
    "log10(Coefficient of variation)")

### OXYGEN CV VERSUS EXPERIMENTAL CONDITIONS

complete.forecasts.pe.cv2.oxygen <- complete.forecasts %>%
  ungroup() %>%
  dplyr::filter(Target == "Mean_oxygen", forecast.method == "Simplex EDM") %>%
  dplyr::select(
    bottle,
    median_richness,
    richness,
    light_treatment,
    composition,
    cv.res,
    median_richness_center,
    incubator
  ) %>%
  dplyr::mutate(log10.cv.res = log10(cv.res))

m.cv.oxy <-
  lmer(
    log10.cv.res ~ median_richness_center * light_treatment +
      (1 | composition) + (1 | incubator),
    complete.forecasts.pe.cv2.oxygen
  )

```

```

newdat.m.cv.oxy <- with(
  complete.forecasts.pe.cv2.oxygen,
  expand.grid(
    median_richness_center = seq(
      min(median_richness_center),
      max(median_richness_center),
      length.out = 100
    ),
    light_treatment = unique(light_treatment)
  )
)

newdat.m.cv.oxy <-
  conf_interval(m.cv.oxy,
    newdat.m.cv.oxy,
    "log10.cv.res",
    "log10(Coefficient of variation)")

##### PLOT

complete.forecasts.pe.cv2.biomass %>%
  ggplot(aes(median_richness_center, log10.cv.res, col = light_treatment)) +
  geom_ribbon(
    data = newdat.m.cv.bio,
    aes(
      ymin = lower,
      ymax = upper,
      y = fit,
      group = light_treatment
    ),
    col = NA,
    fill = "grey",
    alpha = 0.3
  ) +
  geom_line(data = newdat.m.cv.bio,
    size = 1.3,
    aes(y = fit, group = light_treatment)) +
  geom_point(size = 2.5, alpha = 0.6) +
  labs(x = "Centered median realized species richness",
    y = "log10(Coefficient of variation of total biomass)",
    col = "Light conditions") +
  scale_color_manual(values = c("#C4961A", "#293352"),
    aesthetics = c("color", "fill")) +
  theme_bw() +

complete.forecasts.pe.cv2.oxygen %>%
  ggplot(aes(median_richness_center, log10.cv.res, col = light_treatment)) +
  geom_ribbon(
    data = newdat.m.cv.oxy,
    aes(

```

```

    ymin = lower,
    ymax = upper,
    y = fit,
    group = light_treatment
  ),
  col = NA,
  fill = "grey",
  alpha = 0.3
) +
geom_line(data = newdat.m.cv.oxy,
          size = 1.3,
          aes(y = fit, group = light_treatment)) +
geom_point(size = 2.5, alpha = 0.6) +
labs(x = "Centered median realized species richness",
     y = "log10(Coefficient of variation of mean oxygen)",
     col = "Light conditions") +
scale_color_manual(values = c("#C4961A", "#293352"),
                   aesthetics = c("color", "fill")) +
theme_bw() +

plot_layout(guides = "collect") + plot_annotation(tag_levels = "a")

```

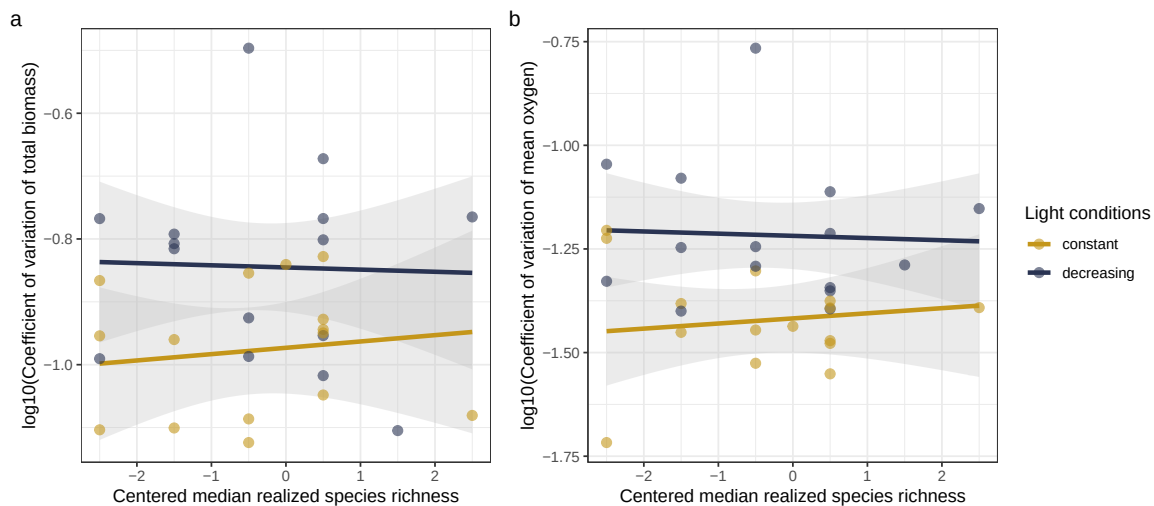


Figure 23: Coefficient of variation of (a) total community biomass, and (b) oxygen concentration. Lines display the fit of the corresponding linear mixed model, shaded areas denote the 95% confidence intervals. Incubator and composition were included as random effects in the model.

```

# compare average cv of species biomass time series with cv of total biomass
# calculate average cv.res of species biomass (of species used as forecast targets)

density.complete.pe.cv <- dd_list$densities.int.detrend.complete %>%
  group_by(
    species,
    bottle,
    composition,
    ForecastTarget,

```

```

richness,
incubator,
method,
light_treatment,
realized_richness,
mean_richness,
median_richness
) %>%
summarize(
  density.pe.5 = PE(
    x = density,
    weighted = T,
    word_length = 5,
    tau = 1,
    tie_method = "noise",
    noise_amount = 10
  ),
  biomass.pe.5 = ifelse(
    all(is.na(biomass)),
    NA,
    PE(
      x = biomass,
      weighted = T,
      word_length = 5,
      tau = 1,
      tie_method = "noise",
      noise_amount = 10
    )
  ),
  biomass.cv.res = sd(biomass.detrended.unstandardized, na.rm =
    T) / mean(biomass.original, na.rm = T),
  density.cv.res = sd(density.detrended.unstandardized, na.rm =
    T) / mean(density.original, na.rm = T),
  biomass.acf_lag5 = stats::acf(
    biomass,
    lag.max = 5,
    plot = F,
    na.action = na.pass
  )$acf[6],
  density.acf_lag5 = stats::acf(
    density,
    lag.max = 5,
    plot = F,
    na.action = na.pass
  )$acf[6]
)

mean.cv.biomass <- density.complete.pe.cv %>%
  dplyr::filter(ForecastTarget == T) %>%
  group_by(bottle,
    median_richness,
    richness,
    light_treatment,

```

```

      composition) %>%
dplyr::summarize(cv.res = mean(biomass.cv.res, na.rm = T)) %>%
dplyr::mutate(level = "population")

temp.data <- complete.forecasts.pe.cv2.biomass
temp.data$level <- "community"

mean.cv.biomass <- full_join(mean.cv.biomass, temp.data)

mean.cv.biomass %>%
  ggplot(aes(
    x = level,
    y = cv.res,
    colour = as.factor(median_richness)
  )) +
  geom_line(aes(group = bottle)) +
  scale_color_manual(values = unname(okabe())) +
  theme_bw() +
  labs(x = "Level of aggregation", y = "Coefficient of variation",
       col = "Median richness") +
  scale_x_discrete(expand = c(0.2, 0))

```

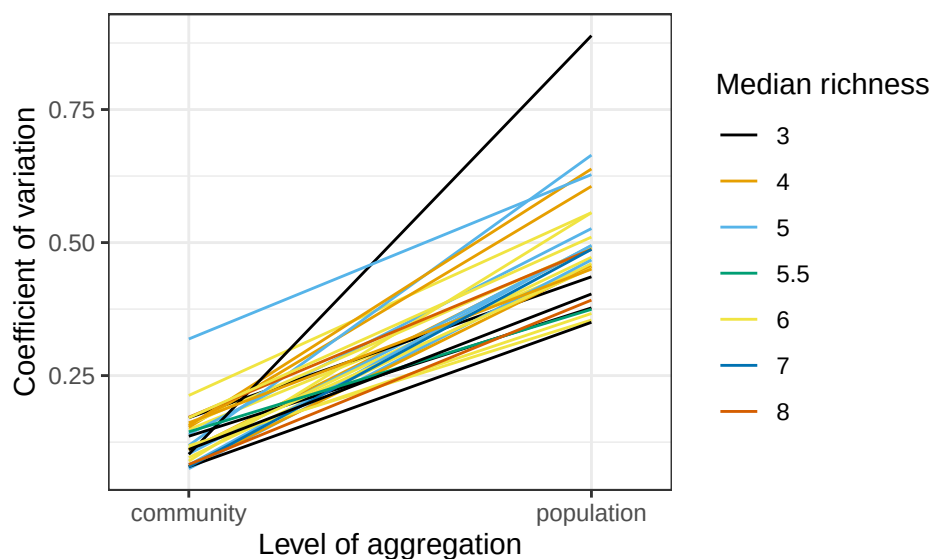


Figure 24: Coefficient of variation (CV) of community and population biomass. For each bottle, we calculated the CV of the time series of total community biomass, and the average CV of the biomass time series of all species in a bottle. Lines connect the two values of a given bottle, and are coloured by median richness.

10 Versions of R and of the R-packages used

```

# Base packages:
s <- sessionInfo()
s1 <- s$basePkgs[1]

```

```

for (i in 2:length(s$basePkgs)) {
  s1 <- paste(s1, ", ", s$basePkgs[i], sep = "")
}

# Other Packages:
pack.info <- installed.packages()
output.packages <- data.frame(pack.info[names(s$otherPkgs),
                                         c("Package", "Version")])

s2 <- paste(names(s$otherPkgs)[1],
            output.packages[names(s$otherPkgs)[1], "Version"])

k <- length(names(s$otherPkgs))

if (k > 1) {
  for (i in 2:k) {
    s2 <- paste(s2, ", ",
                paste(names(s$otherPkgs)[i],
                      output.packages[names(s$otherPkgs)[i], "Version"]),
                sep = "")
  }
}

```

R version: R version 4.3.2 (2023-10-31 ucrt).

Base packages: grid, stats, graphics, grDevices, utils, datasets, methods, base.

Other packages: lavaan.survey 1.1.3, lavaan 0.6-17, survey 4.2-1, survival 3.5-7, broom.mixed 0.2.9.4, GGally 2.2.1, ggh4x 0.2.8, lmerTest 3.1-3, lme4 1.1-35.1, Matrix 1.6-5, here 1.0.1, pals 1.8, kableExtra 1.4.0, data.table 1.15.0, patchwork 1.2.0, lubridate 1.9.3, forcats 1.0.0, stringr 1.5.1, dplyr 1.1.4, purrr 1.0.2, readr 2.1.5, tidyr 1.3.1, tibble 3.2.1, ggplot2 3.4.4, tidyverse 2.0.0, ggstance 0.3.6, emmeans 1.10.0, knitr 1.45.

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